

.lef file

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What is a .lef File ?

Physical view of standard cells and technology for place and route

Tells the tool "Where things are" in the physical world !

LEF = Physical View
= Enabling PnR

1 What is a .lef file ?

- A .lef file is an ASCII text file that describes the **physical geometry** of standard cells, macros and technology information required by **place and route tools**.
- It does **NOT** contain timing or power information (that is in .lib).

2 What is the full form of LEF ?

- LEF stands for **Library Exchange Format**.
- It is an industry standard format (maintained by Si2 and widely used in the ASIC industry).

3 Why is LEF required in ASIC physical design ?

- PnR tools need to know the **physical** information of cells and layers to perform floorplanning, placement, routing and signoff.
- LEF provides cell boundaries, pin locations, routing obstructions, metal layer rules, vias, tracks, etc.
- Without this information, the tool does not know **how** big a cell is, where the pins are, which layers exist, and what the design rules are.

4 Who provides the LEF file ?

- The LEF files are provided by the **foundry** and **standard cell library vendors** (e.g., TSMC, Samsung, GlobalFoundries, ARM, Synopsys, Cadence, etc.).
- For third-party IPs (e.g., SRAM, PLL, CPU), the IP vendor provides the LEF.

5 Why can't Innovus perform PnR without LEF ?

- Innovus needs LEF to understand the physical world :
 - Metal layers and design rules (from technology LEF)
 - Cell sizes, pin locations and obstructions (from cell LEF)
 - Routing tracks, vias, spacing, etc.
- Without LEF, Innovus cannot place cells, route wires or check design rule compliance, because it has no physical geometry information.

LEF vs LIB : Different Information

.lib (Logical View)	.lef (Physical View)
- Timing (cell delay, transition) - Power (dynamic, leakage) - Electrical constraints - Cell functionality (boolean)	- Cell physical size (width, height) - Pin locations and metal layers - Routing obstructions (OBS) - Technology layer rules (for tech.lef) - Vias, tracks, spacing, etc.

How LEF is Used in the ASIC Flow

Design Inputs: RTL (.v), Library (.lib), Physical (.lef) → **cadence Place & Route (Innovus)** → Outputs: Def file (.def), GDSII (.gds), Reports (timing, DRC, etc.)

Example: Small Portion of a LEF File

```

VERSION 5.8 ;
DIVIDERCHAR "/" ;
BUSBITCHARS "[ ]" ;

MACRO AND2_X1
CLASS CORE ;
ORIGIN 0 0 ;
SIZE 1.234 BY 0.720 ;
SYMMETRY X Y ;
SITE CORE ;
...
END AND2_X1
  
```

- File version and basic settings
- Cell name and class
- Physical size (in microns)
- Placement information

Key Takeaway:

- LEF tells the PnR tool "where things are" in the physical world.
- It is essential for successful placement, routing and signoff in ASIC design.

"No LEF, No Layout !"

Why LEF Uses an Abstract View ?

Just enough information for place and route – nothing more, nothing less !

From complex transistors to a simple physical model !

1 What is an abstract view ?

- An abstract view is a **simplified representation** of a cell's physical layout that contains only the information required by place and route tools.
- It hides the detailed transistor-level geometry and keeps only the **essential physical information** such as cell boundary, pin locations and routing obstructions.
- It is generated from the full layout (GDS) using abstraction tools (e.g., Cadence Abstract Generator, Synopsys Milkyway).

Full Layout (GDS) – Detailed View

Example: AND2_X1 (Full GDS layout)
Contains every transistor, wire, contact, etc.
Size: ~ 50 KB (for one cell)

Abstract View (LEF) – Simplified View

Example: AND2_X1 (LEF abstract view)
Contains only cell boundary, pin locations and OBS.
Size: ~ 2 KB (for one cell)

Abstraction Tool
(Removes unnecessary details)

2 What information is removed from GDS ?

- Transistor-level details (NMOS, PMOS, etc.)
- Exact shapes of diffusion, poly, implants
- Internal wires and connections
- Detailed layer geometries not needed for routing (e.g., active, well, implant layers)
- Device parameters and electrical data (these are in .lib, not in .lef)

3 What information does LEF keep ?

- Cell boundary (PR boundary)
- Pin locations with layer and exact coordinates
- Routing obstructions (OBS)
- Cell size (width and height)
- Placement information (SITE, symmetry, class)
- Meta information required by PnR tools

LEF keeps only what the router needs to place cells, connect pins and avoid blockages.

4 Why can't PnR directly use full GDS ?

- GDS is **too large** (contains every layer and every device).
- Parsing full GDS is very slow and memory intensive.
- PnR tools do not need transistor-level details.
- GDS contains information that is irrelevant for placement and routing (e.g., diffusion, implants, exact device shapes).
- Using full GDS would expose IP details (no IP protection).
- It would make the PnR flow extremely slow and inefficient.

5 How does abstraction improve tool performance ?

Parameter	Full GDS (Detailed)	LEF (Abstract)
File size (AND2_X1)	~ 50 KB	~ 2 KB
Contains transistors	Yes	No
Contains internal wires	Yes	No
Used by PnR	No (too large, slow)	Yes
IP protection	No	Yes
Read time in tool	Very slow	Fast

Key Point:
LEF is not a replacement for GDS. It is a simplified, router-focused view of the layout that makes PnR fast, efficient and secure.

Takeaway:

Abstraction removes the unnecessary details and keeps only what is required for physical implementation, making the place and route flow faster, simpler and more efficient.

Right information for the right tool !

GDS vs LEF

Same design, different purpose – One for manufacturing, one for place & route.

GDS is the complete truth, LEF is the useful summary !

1 What is GDS ?

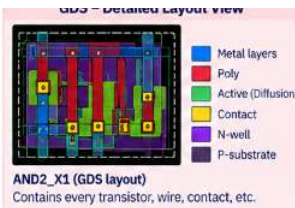
2 What is LEF ?

3 Visual Comparison (Same Cell: AND2_X1)

GDS – Detailed Layout View | LEF – Abstract View

- GDS (GDSII) is a **binary file format** that contains the **complete physical layout** of a design.
- It includes every layer, every shape and every device (transistors, diffusions, polys, metals, vias, etc.).
- It is used for **manufacturing** (fabrication) at the foundry.

- LEF (Library Exchange Format) is an **ASCII text file** that describes the **abstract physical view** of cells, macros and technology.
- It contains only the information required by place and route tools (cell boundary, pin locations, routing obstructions, layer rules, etc.).
- It is used for **physical implementation (PnR)**, not for manufacturing.



4 Key Differences Between GDS and LEF

Parameter	GDS (GDSII)	LEF (Library Exchange Format)
File format	Binary	ASCII (text)
Purpose	Manufacturing (fab)	Place and Route (PnR)
Content level	Full transistor-level geometry	Abstract physical view
Information	All layers, shapes, devices	Cell boundary, pins, OBS, layer rules
File size (per cell)	Large (~50 KB for a small cell)	Small (~2 KB for the same cell)
Readability	Not human readable	Human readable (can be opened in text editor)
Used by	Foundry (tape-out)	PnR tools (Innovus, ICC2, etc.)
Contains device details?	Yes	No
Used for signoff?	Yes (for manufacturing)	No
Example tools	Virtuoso, KLayout (view)	Text editor, Innovus, ICC2 (read)

5 Answers to the Questions

- What is GDS?**
GDS (GDSII) is a binary file that contains the complete physical layout of a design, used for manufacturing.
- What is LEF?**
LEF is an ASCII text file that contains the abstract physical view of cells, macros and technology, used for place and route tools.
- What is the difference between GDS and LEF?**
GDS has the full detailed layout for manufacturing, while LEF has a simplified abstract view with only the necessary information for PnR.
- Which one contains full transistor-level geometry?**
GDS contains full transistor-level geometry.
- Which one is used by PnR?**
LEF is used by PnR tools (e.g., Cadence Innovus, Synopsys ICC2).



Key Takeaways:

- GDS is the **complete layout used for fabrication**.
- LEF is a **simplified abstract view used for place and route**.
- Same design, different purpose** – both are required in the ASIC flow.



Right view for the right step in the ASIC flow!



Technology LEF

Defines the process technology – metal layers, vias, tracks and design rules for the whole chip.

No technology LEF, No routing!

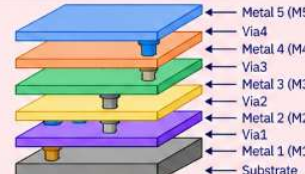
1 What is Tech LEF?

- Tech LEF (Technology LEF) is an ASCII text file that describes the technology information required by place and route tools.
- It defines the **physical rules of the process** such as metal layers, via definitions, routing tracks, spacing rules, design rules, etc.
- It is not cell-specific – it covers the **entire technology node**.

2 What information does Tech LEF contain?

- Layer definitions:** metal layers (M1, M2, ...), cut/via layers (VIA1, VIA2, ...), implant layers, etc.
- Routing rules:** track pitch, direction, width, spacing, etc.
- Via definitions and via rules**
- Design rule information** (spacing tables, min area, etc.)
- Electrical properties:** resistance, capacitance (optional)
- Manufacturing grid**
- Technology-specific statements** (e.g., UNITS, VERSION)

Example: Metal Stack (from Tech LEF)



3 Does Tech LEF describe individual cells?

- No**, Tech LEF does **NOT** describe individual cells.
- It does not contain cell geometries, pins or obstructions.
- It only defines the technology rules and layer information that apply to the **whole chip**.
- Individual cell information (size, pins, OBS, etc.) is provided in the **Cell LEF** (stdcell.lef, macro.lef).

4 Why is Tech LEF required before Cell LEF?

- The PnR tool must first **understand the technology** (layers, design rules, vias, etc.) before it can read and interpret the cell LEF.
- Cell LEF **references layer names** (e.g., M1, M2, VIA1) defined in Tech LEF.
- If Tech LEF is not loaded first, the tool will not know what these layers mean and will give errors (e.g., "undefined layer").

5 Who provides Tech LEF?

- Tech LEF is provided by the **foundry** for a specific technology node (e.g., TSMC 5nm, Samsung 7nm, GlobalFoundries, etc.).
- It may also be provided by the **standard cell library vendor** (e.g., ARM, Synopsys, Cadence) along with the PDK.
- There is typically one Tech LEF file per technology node.

Small Example Snippet from a Technology LEF

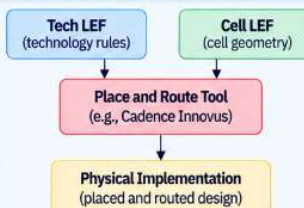
```

VERSION 5.8 ;
BUSBITCHARS "[ ]" ;
DIVIDERCHAR "/" ;
UNITS DISTANCE MICRONS 1000 ; # 1 unit = 0.001 micron
MANUFACTURINGGRID 0.005 ;

LAYER M1
TYPE ROUTING ;
DIRECTION HORIZONTAL ;
PITCH 0.040 ;
WIDTH 0.020 ;
SPACING 0.020 ;
END M1

```

Where Tech LEF Fits in the PnR Flow



File Naming Example

```

tsmc_5nm_tech.lef → Technology LEF
tsmc_5nm_stdcell.lef → Cell LEF (standard cells)
tsmc_5nm_macro.lef → Cell LEF (macros)

```

Key Takeaways

- Tech LEF defines the process, not the cells.
- It contains layer, via, routing and design rule information.
- It must be loaded before Cell LEF in PnR tools.
- Provided by foundry or library vendor.
- One Tech LEF per technology node.

Defines the Rules of the Game!



Tech LEF Header and Units

Basic settings at the top of a Technology LEF file that tell the tool how to read the file.

Small Settings... Big Impact!

tsmc_5nm_tech.lef (excerpt)

```

1  VERSION 5.8 ;
2  DIVIDERCHAR "/" ;
3  BUSBITCHARS "[ ]" ;
4  DATABASE MICRONS 1000 ;
5  MANUFACTURINGGRID 0.001 ;
6
7  LAYER M1
8    TYPE ROUTING ;
9    DIRECTION HORIZONTAL ;
10   PITCH 0.040 ;
11   WIDTH 0.020 ;
12   SPACING 0.020 ;
13   END M1

```

1 What is `VERSION`?

- Specifies the LEF file format version.
- Indicates which LEF syntax and features are supported.
- Common versions: 5.6, 5.7, 5.8 (latest).

Example:

```

VERSION 5.8 ;
→ This file follows LEF 5.8 syntax.

```

2 What is `DIVIDERCHAR`?

- Specifies the character used to separate hierarchy levels in cell names.
- Commonly "/" (slash).
- Helps the tool understand hierarchical names.

Example:

```

DIVIDERCHAR "/" ;
→ Cell name like
top/u1/u2/AND2_X1

```

3 What is `BUSBITCHARS`?

- Specifies the characters used to denote bus bits.
- Typically uses "[]" (square brackets).
- Helps the tool recognize bus names like data[7:0].

Example:

```

BUSBITCHARS "[ ]" ;
→ Bus name: data[7:0]

```

4 What is 'DATABASE MICRONS' ?

Example:
DATABASE MICRONS 1000 ;
→ 1 μm = 1000 database units (DBU)
So, 0.001 μm = 1 DBU (1 nm)

- Defines the units used in the LEF file.
- It specifies how many database units (DBU) are there in 1 micron (μm).
- All dimensions in the LEF are integer values in DBU.

5 Why are LEF units important ?

- All physical dimensions (layer widths, spacing, coordinates, etc.) are defined in database units (DBU).
- Ensures consistent and accurate geometry interpretation across all tools (PnR, DRC, extraction).
- If units are wrong, the entire design can be scaled incorrectly (e.g., 10 $\times$ bigger or smaller).
- Helps maintain precision for advanced nodes (e.g., 5nm, 3nm).

Real Example: Unit Conversion

If DATABASE MICRONS 1000 ;
 1 μm = 1000 DBU
 0.1 μm = 100 DBU
 0.01 μm = 10 DBU
 0.001 μm = 1 DBU (1 nm)
 In the LEF file, dimensions are written in DBU (integers).

Key Takeaway

- The LEF header defines how the tool reads and interprets the file.
- VERSION, DIVIDERCHAR, BUSBITCHARS and DATABASE MICRONS are basic but essential settings.
- Correct units ensure accurate physical implementation.
- Always check the header and units before using a LEF file.

Correct Units = Correct Chip !

Reference: Synopsys LEF/DEF Reference Manual | Cadence Innovus User Guide | TSMC/Samsung Foundry Documentation

Read the Header, Avoid the Error !

Manufacturing Grid in LEF

All geometries in LEF must align to a fixed grid for accurate manufacturing and implementation.

Everything snaps to the grid !

1 What is 'MANUFACTURINGGRID' ?

- 'MANUFACTURINGGRID' defines the basic grid resolution for all physical geometries in the LEF file.
- It specifies the **smallest distance** (in microns) to which all **coordinates** (layer shapes, pins, vias, etc.) must align.

Example: MANUFACTURINGGRID 0.005 ;
means the grid is 0.005 μm (5 nm).

Example in LEF

```
VERSION 5.8 ;
DIVIDERCHAR "/" ;
BUSBITCHARS "[" ;

UNITS DISTANCE MICRONS 1000 ;
MANUFACTURINGGRID 0.005 ;
...
```

Grid = 0.005 μm (5 nm)

All coordinates must be multiples of 0.005 μm (e.g., 0.000, 0.005, 0.010, ...)

Grid Visualization

On-grid (Valid)

All coordinates are multiples of 0.005 μm

2 Why is a manufacturing grid required ?

- Ensures all geometries align to a **common, fixed grid**.
- Matches the **manufacturing process resolution** (lithography).
- Avoids **precision and rounding errors**.
- Keeps **consistency** between technology LEF, cell LEF, and design tools.
- Helps tools like Innovus, DRC and LVS **interpret geometry correctly**.

3 What happens if geometry is off-grid ?

- Tool will give **errors or warnings** (e.g., "not on grid").
- Shapes may be **rounded** to nearest grid point, changing the intended size/position.
- Can cause **DRC violations**.
- May lead to **misalignment** between layers (e.g., via, metal).
- Can create **connectivity issues** or manufacturing problems.

Off-Grid Example (Invalid)

(0.012, 0.017) Not a multiple of 0.005

(0.003, 0.007) Not a multiple of 0.005

Off-grid (Invalid)

4 How does the manufacturing grid affect placement and routing ?

- All **cell boundaries**, **pin locations**, and **routing shapes** must be on the grid.
- Placement:** Standard cell rows, site dimensions, and cell origins are defined using grid multiples.
- Routing:** Tracks, wire widths, spacings and via locations are defined on the grid.
- Ensures that all tools (PnR, DRC, extraction) use the **same coordinate system**.
- Helps in achieving **accurate, manufacturable and reliable designs**.

Example: Placement Grid

Cell width = $N \times \text{grid}$
(e.g., $20 \times 0.005 = 0.1 \mu\text{m}$)

Example: Routing Track Grid

Routing tracks are placed on grid multiples

Track pitch = $N \times \text{grid}$
(e.g., $0.04 \mu\text{m} = 8 \times 0.005$)

Key Takeaway:

The **manufacturing grid** defines the smallest allowed step for all geometries in LEF. Keeping everything on grid ensures correct placement, routing and successful manufacturing.

Align to the Grid Design for Reality !

Layer Types in LEF

Different layer types serve different purposes in the physical design flow.

Layers define the routing world !

1 What is a 'ROUTING' layer ?

- A ROUTING layer is used for signal routing (wires).
- It includes metal layers (e.g., M1, M2, ...).
- Defines properties like direction, pitch, width, spacing, resistance, capacitance, etc.
- Used to route nets during place and route.

Example (LEF):

```
LAYER M2
TYPE ROUTING ;
DIRECTION VERTICAL ;
PITCH 0.040 ;
WIDTH 0.020 ;
SPACING 0.020 ;
END M2
```

2 What is a 'CUT' layer ?

- A CUT layer is used for vias (contacts) between two metal layers.
- Represents the via shape (e.g., VIA1, VIA2, ...).
- Defines via size, spacing, enclosure rules, etc.
- Used to connect one metal layer to another.

Example (LEF):

```
LAYER VIA1
TYPE CUT ;
WIDTH 0.060 ;
HEIGHT 0.060 ;
SPACING 0.060 ;
END VIA1
```

3 What is a 'MASTERSLICE' layer ?

- A MASTERSLICE layer is used to define the placement area for standard cells (the rows).
- It defines the vertical extent of the standard cell site/row.
- Usually defined only in technology LEF.
- Helps the tool understand where cells can be placed.

Example (LEF):

```
LAYER MASTERSLICE
TYPE MASTERSLICE ;
WIDTH 0.272 ; # example height
END MASTERSLICE
```

4 Difference between metal and via layers ?

- Metal layers (ROUTING)** are used to route wires horizontally or vertically.
- Via layers (CUT)** are used to connect two metal layers.
- Metal layers carry signals, vias provide vertical connectivity.
- Metals have direction, pitch, width, etc., while vias have cut size and enclosure rules.

Visual Example: Layer Stack

← Metal 3 (M3)
← VIA2 (CUT)
← Metal 2 (M2)
← VIA1 (CUT)
← Metal 1 (M1)
← Substrate

Top View (Layout View)

Legend:
 Metal 2 (ROUTING)
 Metal 1 (ROUTING)
 Via1 (CUT)

Comparison Table

Feature	Metal Layer (ROUTING)	Via Layer (CUT)	Masterslice (MASTERSLICE)
Type	ROUTING	CUT	MASTERSLICE
Purpose	Signal routing	Connects two metals	Defines cell placement rows
Examples	M1, M2, M3, ...	VIA1, VIA2, ...	MASTERSLICE
Direction	Horizontal/Vertical	Not applicable	Not applicable
Key Parameters	Pitch, width, spacing, resistance	Cut size, spacing, enclosure	Width (row height)
Carries signal?	Yes	No (only connection)	No

Key Takeaways:

- ROUTING** layers are for wires, **CUT** layers are for vias, and **MASTERSLICE** defines placement rows.
- Metal layers** route signals, **via layers** connect metals, and **masterslice** enables standard cell placement.
- Each layer type has different properties defined in the technology LEF.

Right layer Right connection Right design !

Routing Layer Definition in LEF

Defines the physical and electrical properties of a metal routing layer.

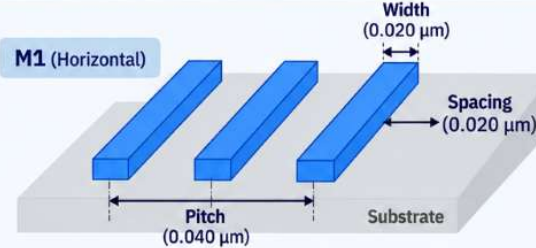
These parameters guide the router!

Example: M1 Layer Definition in Tech LEF

```
1 LAYER M1
2 TYPE ROUTING ;
3 DIRECTION HORIZONTAL ;
4 PITCH 0.040 ;
5 WIDTH 0.020 ;
6 SPACING 0.020 ;
7 RESISTANCE RPERSQ 0.075 ;
8 CAPACITANCE CPERSQDIST 0.20 ;
9 END M1
```

(Tech LEF snippet)

Visual View of M1 Parameters



1 What is 'LAYER M1' ?

- 'LAYER M1' defines the first metal routing layer in the technology.
- Each layer in LEF has a unique name (M1, M2, ...).
- It specifies the physical and electrical properties of that layer.
- It is used for signal routing during place and route.

Example: LAYER M1
→ Defines metal layer 1

2 What is 'DIRECTION' ?

- Specifies the preferred routing direction for the layer.
- Usually, odd metal layers (M1, M3, ...) are HORIZONTAL.
- Even metal layers (M2, M4, ...) are VERTICAL.
- Helps in reducing congestion and improving routability.



3 What is 'PITCH' ?

- Distance between the center of two adjacent routing tracks.
- It defines the routing grid for the layer.
- Pitch = Width + Spacing (in simple cases).
- All routing must follow the pitch grid.



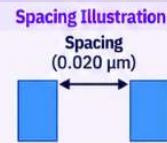
4 What is 'WIDTH' ?

- Minimum allowed width of a wire on this layer.
- Defines the thickness of the routing wire.
- Tool can use wider wires if needed (for low resistance or high current).



5 What is 'SPACING' ?

- Minimum allowed spacing between two adjacent wires on the same layer.
- Prevents short-circuit and manufacturing defects.
- Spacing rules may vary for different widths or special cases (e.g., parallel run length).



Key Takeaways.

- 'LAYER M1' defines the first metal routing layer.
- 'DIRECTION' sets the preferred routing direction.
- 'PITCH' is the track-to-track distance.
- 'WIDTH' is the minimum wire width.
- 'SPACING' is the minimum gap between wires.
- These parameters ensure correct, manufacturable and routable designs.

Right Rules
Enable
Reliable Chips!



Routing Direction and Track Grid in LEF

Preferred direction and track grid help in efficient and manufacturable routing.

Follow the Grid
Route with Confidence!

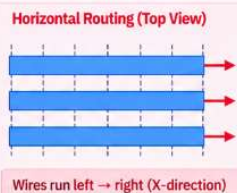
1 Why does a routing layer have a preferred direction ?

- Each metal layer is given a preferred routing direction (horizontal or vertical) to reduce routing congestion.
- Adjacent metal layers usually have orthogonal directions (e.g., M1: horizontal, M2: vertical).
- It helps in better wire distribution, improves routability and follows manufacturing practices.

Example from LEF:
LAYER M1
TYPE ROUTING ;
DIRECTION HORIZONTAL ;
PITCH 0.040 ;
END M1

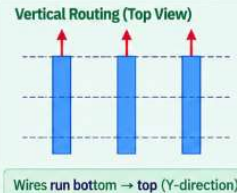
2 What is horizontal routing ?

- Routing in the left-to-right direction (along X-axis) is called horizontal routing.
- Usually used on layers with DIRECTION HORIZONTAL (e.g., M1, M3).
- Wires run parallel to the X-axis.



3 What is vertical routing ?

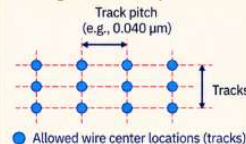
- Routing in the bottom-to-top direction (along Y-axis) is called vertical routing.
- Usually used on layers with DIRECTION VERTICAL (e.g., M2, M4).
- Wires run parallel to the Y-axis.



4 What is a routing track ?

- A routing track is an imaginary line on which the center of a wire can be placed.
- Tracks are defined based on the PITCH value in LEF.
- All routing must follow the track grid to ensure DRC clean and manufacturable design.

Routing Track Grid (Top View)



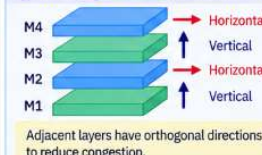
5 How do pitch and direction affect routing congestion ?

- Smaller pitch (more tracks) → more routing resources → less congestion.
- Larger pitch (fewer tracks) → fewer routing resources → higher congestion.
- If too many nets need to route in the same direction on a layer, congestion increases.
- Using alternate directions in adjacent layers (horizontal/vertical) helps distribute routing and reduces congestion.
- Proper track planning and layer usage is critical for meeting timing and routability.

Effect of Pitch on Number of Tracks



Typical Routing Directions in Advanced Nodes



Key Takeaways:

- Preferred direction improves routability and reduces congestion.
- A routing track is defined by the pitch and acts as the allowed wire center location.
- Smaller pitch gives more tracks, larger pitch increases congestion.
- Adjacent layers (M1, M2, ...) usually have orthogonal directions (horizontal/vertical).

Right Direction
Right Tracks
Better Routing!



Metal Electrical Properties in LEF

These parameters model the resistance and capacitance of metal wires for accurate timing and parasitic extraction.

Real Numbers
Real Performance!

1 What is 'RESISTANCE RPERSQ' ?

- 'RESISTANCE RPERSQ' defines the sheet resistance of the metal layer (in ohms per square, Ω/□).



2 What is 'CAPACITANCE CPERSQDIST' ?

- 'CAPACITANCE CPERSQDIST' defines the coupling capacitance between two parallel metal wires per unit length.



1 What is 'RESISTANCE R_{PERSQ}' ?

- 'RESISTANCE R_{PERSQ}' defines the sheet resistance of the metal layer (in ohms per square, $\Omega/\square$).
- It gives the resistance of a square piece of metal (length = width).
- Used by tools to calculate the resistance of wires of any length and width.

Wire Resistance: $R = R_{PERSQ} \times \left(\frac{L}{W}\right)$

R_{PERSQ} = resistance per square ($\Omega/\square$)
 L = wire length
 W = wire width

If $R_{PERSQ} = 0.075 \Omega/\square$,
 $L = 10 \mu\text{m}$, $W = 0.5 \mu\text{m}$
 $R = 0.075 \times (10 / 0.5) = 1.5 \Omega$

2 What is 'CAPACITANCE C_{PERSQDIST}' ?

Capacitance between two parallel metal wires per unit area and per unit distance ($\text{fF}/\mu\text{m}^2/\mu\text{m}$).

- It models the side (lateral) coupling capacitance between neighboring wires on the same layer.
- Used to calculate the coupling capacitance based on overlap area and spacing.

Coupling Capacitance: $C_{\text{couple}} = C_{PERSQDIST} \times \left(\frac{A}{D}\right)$

$C_{PERSQDIST}$ = coupling capacitance per square distance ($\text{fF}/\mu\text{m}^2/\mu\text{m}$)
 A = overlapping area between wires (μm^2)
 D = spacing between wires (μm)

If $C_{PERSQDIST} = 0.20 \text{ fF}/\mu\text{m}^2/\mu\text{m}$
 $L = 10 \mu\text{m}$, $W = 0.5 \mu\text{m}$, $D = 0.2 \mu\text{m}$
 $A = 10 \times 0.5 = 5 \mu\text{m}^2$
 $C_{\text{couple}} = 0.20 \times (5 / 0.2) = 5 \text{ fF}$

3 What is 'EDGE CAPACITANCE' ?

- 'EDGE CAPACITANCE' defines the fringe capacitance between the edge of a metal wire and the substrate or surroundings ($\text{fF}/\mu\text{m}$).
- It accounts for the capacitance from the side edges of the wire to the substrate (or other reference).
- It is independent of coupling to other wires.

Metall wire (top view)

Edge capacitance (on both sides)

Substrate

If EDGE CAPACITANCE = $0.02 \text{ fF}/\mu\text{m}$ and wire length $L = 10 \mu\text{m}$
 $C_{\text{edge (total)}} = 2 \times 0.02 \times 10 = 0.4 \text{ fF}$ (2 edges)

4 How do metal R and C affect timing ?

- Metal resistance (R) and capacitance (C) form an RC network that causes signal delay (propagation delay).
- Higher R $\rightarrow$ larger voltage drop and slower signal.
- Higher C $\rightarrow$ more charge to drive, slower signal.
- Together, they increase the RC delay, which affects setup/hold timing and overall performance.

Driver $\rightarrow$ R (metal) $\rightarrow$ Load

C (metal + coupling + edge)

Delay (simplified): $t_{\text{delay}} \approx 0.69 \times R \times C$

Higher R or C $\rightarrow$ Higher delay $\rightarrow$ Timing violations possible

5 How are these properties related to parasitic extraction ?

- The values in LEF (R_{PERSQ}, C_{PERSQDIST}, EDGE CAPACITANCE) are used by parasitic extraction tools (e.g., SPEF generation) to calculate the actual R and C of each net based on the metal geometry.
- Extraction tools use the layer definitions and these electrical parameters to model wire resistance and capacitance accurately.
- The extracted parasitics (in SPEF) are then used for timing analysis (STA) to verify performance.

LEF (metal R, C) $\rightarrow$ Parasitic Extraction (PEX) $\rightarrow$ SPEF (net R, C) $\rightarrow$ Timing Analysis (STA)

Accurate Models for a Faster, Reliable Chip!

Key Takeaways:

- R_{PERSQ} models wire resistance, C_{PERSQDIST} models coupling capacitance, and EDGE CAPACITANCE models fringe capacitance.
- These parameters are essential for accurate parasitic extraction and timing analysis.
- Correct metal R and C values ensure reliable performance and timing closure in ASIC designs.

Metal Electrical Properties in LEF

These parameters model the resistance and capacitance of metal wires for accurate timing and parasitic extraction.

1 What is 'RESISTANCE R_{PERSQ}' ?

- 'RESISTANCE R_{PERSQ}' defines the sheet resistance of the metal layer (in ohms per square, $\Omega/\square$).
- It gives the resistance of a square piece of metal (length = width).
- Used by tools to calculate the resistance of wires of any length and width.

Wire Resistance: $R = R_{PERSQ} \times \left(\frac{L}{W}\right)$

R_{PERSQ} = resistance per square ($\Omega/\square$)
 L = wire length
 W = wire width

If $R_{PERSQ} = 0.075 \Omega/\square$,
 $L = 10 \mu\text{m}$, $W = 0.5 \mu\text{m}$
 $R = 0.075 \times (10 / 0.5) = 1.5 \Omega$

2 What is 'CAPACITANCE C_{PERSQDIST}' ?

Capacitance between two parallel metal wires per unit area and per unit distance ($\text{fF}/\mu\text{m}^2/\mu\text{m}$).

- It models the side (lateral) coupling capacitance between neighboring wires on the same layer.
- Used to calculate the coupling capacitance based on overlap area and spacing.

Coupling Capacitance: $C_{\text{couple}} = C_{PERSQDIST} \times \left(\frac{A}{D}\right)$

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Metal Spacing Rules in LEF

Spacing rules ensure reliable, manufacturable and high-yield designs.

1 What is minimum spacing ?

- Minimum spacing is the smallest allowed distance between two metal wires on the same layer.
- It prevents short-circuit, reduces coupling and ensures manufacturability.
- Defined in LEF using the SPACING parameter (or a spacing table).

Minimum Spacing (S)

Substrate

2 What is a spacing table ?

- A spacing table defines the required spacing as a function of wire width, parallel run length (or both).
- Used because a single fixed spacing may not be sufficient for all cases.
- LEF supports different table types, e.g.,
 - WIDTH spacing table (spacing vs. width)
 - PARALLEL RUN LENGTH spacing table (spacing vs. run length)
 - TWO WIDTH / TWO-SPACING table (spacing vs. both widths)

Example: Simple WIDTH Spacing Table (LEF)

Wire Width (μm)	Minimum Spacing (μm)
0.04	0.04
0.08	0.06
0.16	0.08
0.32	0.12

3 What is Parallel Run Length (PRL) ?

- Parallel Run Length** is the length over which two adjacent wires run parallel to each other within a certain spacing.
- Longer parallel run length increases coupling (crosstalk), so a larger spacing is required.
- LEF can specify spacing rules based on PRL.

Parallel Run Length (PRL)

Substrate

4 Why does required spacing increase for wider wires ?

- Wider wires have larger edge area, which increases fringe capacitance to neighboring wires.
- Higher coupling can cause more crosstalk and noise.
- To keep coupling within limits, larger spacing is required for wider wires.
- This helps maintain signal integrity and reliable operation.

Narrow wires Medium wires Wide wires

5 How does Innovus use spacing rules ?

- Innovus reads spacing rules from the LEF file (including spacing tables).
- During placement and routing, it checks that all wires and shapes satisfy the required spacing based on width, parallel run length, and layer rules.
- If a violation occurs, Innovus will either:
 - Increase the spacing,
 - Reroute the net, or
 - Report a DRC (Design Rule Check) violation.
- This ensures the final layout is DRC clean and manufacturable.



Via and Via Rules in LEF

Vias connect metal layers and their rules ensure reliable vertical connectivity.

Small Connection Big Impact!

1 What is a via?

- A via is a vertical connection between two metal layers in an IC.
- It provides an electrical path for current to flow from one metal layer to another.
- A via is typically a small conductive plug (e.g., tungsten or copper) that fills a hole and connects the two metals.
- In LEF, vias are defined separately from metal layers.

2 Why is a via required between metal layers?

- Each metal layer is on a different physical level (different height).
- Metals on different layers cannot touch directly.
- A via is required to provide a low-resistance electrical connection between them.
- Without vias, signals could only move in one metal layer, making complex routing impossible.

Vias enable 3D interconnection between metal layers.

3 What is M1-VIA1-M2?

- M1-VIA1-M2 means VIA1 is used to connect Metal 1 (M1) and Metal 2 (M2).
- Each via type in LEF is defined for a specific adjacent metal layer pair (e.g., VIA1: M1 to M2, VIA2: M2 to M3, ...).
- The via sits between the two metals and makes the electrical connection.

M1 - VIA1 - M2 = Metal 1 connected to Metal 2 using VIA1

4 What does a VIA definition contain?

- A VIA definition in LEF contains:
 - Layer names (bottom metal, via layer, top metal)
 - Geometry (size of the via cut)
 - Enclosure rules (how much metal must overlap the via)
 - Spacing rules (distance to other vias)
 - Array/spacing information (for multiple vias)

Example: VIA1 definition (LEF)

```
VIA VIA1
LAYER M1 ;
ENCLOSURE 0.02 ;
LAYER VIA1 ;
RECT 0.0 0.0 0.06 0.06 ;
LAYER M2 ;
ENCLOSURE 0.02 ;
END VIA1
```

Defines geometry and enclosure rules for VIA1 between M1 and M2.

5 What is via enclosure?

- Via enclosure is the amount of metal that must extend beyond the via cut on all sides.
- It ensures good electrical contact, mechanical stability and manufacturability.
- If enclosure is too small, it can lead to high resistance or via failure.

Top enclosure (on M2)
Bottom enclosure (on M1)
Metal must extend beyond the via cut by the specified enclosure on all sides.

Key Takeaways:

- A via is a vertical connection between two metal layers.
- Vias are required because metal layers are on different levels.
- Each via type (e.g., VIA1) connects a specific adjacent metal pair (e.g., M1-M2).
- VIA definitions include geometry, enclosure and spacing rules.
- Via enclosure ensures reliable electrical contact and manufacturability.

Vias Bridge Layers, Enable Great Designs!

Via Rules in LEF

Define via geometry, spacing, enclosure and usage for reliable inter-layer connections.

Connect Layers Design Better!

1 What is 'VIARULE'?

- 'VIARULE' in LEF defines the rules to generate vias between two metal layers.
- It specifies via size, spacing, metal enclosure, and other constraints instead of defining a fixed via shape.
- Used for generated vias (not for pre-defined cut shapes).
- Helps the tool create valid vias for different metal layer pairs.

Example: VIARULE (Tech LEF)

```
VIARULE GENERIC_VIA12
LAYER M1 ;
ENCLOSURE 0.02 ;
LAYER M2 ;
ENCLOSURE 0.02 ;
CUTSIZE 0.06 BY 0.06 ;
SPACING 0.06 ;
END GENERIC_VIA12
```

2 What is generated via?

- A generated via is a via created by the tool (e.g., during placement and routing) based on the 'VIARULE'.
- It is not a fixed, pre-designed via cell, but generated on-the-fly using the rules (size, spacing, enclosure, etc.).
- The router automatically generates these vias wherever needed to connect adjacent metal layers.

The via shape and size are created by the tool based on the VIARULE.

3 What is via spacing?

- Via spacing is the minimum allowed distance between two vias on the same layer.
- It prevents short-circuit and reduces coupling between vias.
- Specified in 'VIARULE' using the SPACING parameter.

VIA1 layer (cut layer)

4 What is metal enclosure?

- Metal enclosure is the amount of metal that must extend beyond the via cut on all sides.
- It ensures good electrical contact, mechanical strength and reliability.
- Specified in 'VIARULE' using the ENCLOSURE parameter for each metal layer (top and bottom).

Metal must extend beyond the via cut by the specified enclosure on all sides.

5 How does the router select vias?

- The router reads the VIARULE definitions from the Tech LEF.
- For each adjacent metal layer pair (e.g., M1-M2), it selects a valid via type based on:
 - Availability of via rules for that layer pair
 - Via size, spacing and enclosure constraints
 - Design rules and current density limits
 - Optimization goals (e.g., minimize resistance, meet timing)
- The router automatically generates and places vias that satisfy all these rules.

Router Flow for Via Selection

```
Read Tech LEF (VIARULE) → Check valid vias for layer pair (e.g., M1-M2) → Verify rules (size, spacing, enclosure, etc.) → Select best via (based on DRC, timing, resistance, etc.) → Generate and place via
```

Key Takeaways:

- 'VIARULE' defines the rules to generate vias between metal layers.
- Generated vias are created by the tool using these rules.
- Via spacing prevents short-circuit and reduces coupling.
- Metal enclosure ensures reliable electrical contact.
- The router selects vias based on VIARULE, DRC rules and optimization goals.

Follow the Rules Build Strong Connections!

Antenna Rules in LEF

Prevent antenna effect and protect gate oxides during manufacturing.

Design Today No Antenna Problems Tomorrow!

1 What is antenna effect?

- Antenna effect occurs during IC manufacturing (e.g., plasma etch).
- Long metal wires can collect charge (like an antenna) from the plasma.
- This charge can flow to a gate through a connected net and damage the thin gate oxide, leading to a gate rupture.

2 Why are antenna rules defined in technology data?

- Different processes and layers have different antenna sensitivities.
- Antenna rules (ratios and limits) are defined in the technology data (LEF/techfile) to:
 - Specify the maximum allowed metal area before a diode or gate is connected.

3 What is antenna area ratio?

- Antenna area ratio is the ratio of metal area (connected to a gate) to the gate area (or diffusion area).
- It indicates how much charge can be collected on the metal during etch relative to what the gate can safely tolerate.
- If the ratio exceeds the technology limit, it is an antenna violation.

net and damage the thin gate oxide, leading to reliability issues or device failure.

Long metal wire (acts like an antenna)

Charge from plasma etch

High voltage can damage gate oxide (antenna effect)

Source Gate Drain Substrate

- Provide layer-wise antenna ratios and diode information.
- Help EDA tools (place & route, signoff) check and prevent antenna violations.

Example (LEF - Antenna Rule)

```
LAYER M1
ANTENNARATIO 400 ; # Max metal/gate ratio
ANTENNADIFFRATIO 200 ;
ANTENNAGATERATIO 300 ;
END M1
```

These values tell the tools the safe limits for each layer.

Antenna Area Ratio = Total Metal Area (connected to gate) / Gate Area (or Diffusion Area)

Metal area = $L \times W$

Gate area = $W_g \times L_g$

Source Gate Drain

4 How can routing create antenna violations ?

- During global/detailed routing, long metal wires are created before reaching a cell (gate or diode).
- These unconnected metal wires (open nets) can collect a large amount of charge during plasma etch.
- If the metal area becomes too large compared to the gate area, it exceeds the antenna ratio limit → antenna violation.

(1) Early routing stage (open net) Long unconnected metal wire Gate (not yet connected)

(2) After more routing Metal area increases

(3) Final connection May violate antenna ratio

5 How are antenna violations fixed ?

- Add antenna diodes (diode to VDD/VSS) to provide a discharge path for the collected charge.
- Use smaller metal segments (insert jogs or connect earlier).
- Use higher metal layers (lower antenna sensitivity).
- Use antenna-aware routing (tool avoids creating violations).
- In some cases, add jumpers or connect to diffusion earlier.

(A) Add antenna diode Antenna diode to VDD/VSS Provides a discharge path for collected charge.

(B) Use smaller segments Break long wires into smaller segments.

(C) Use higher metal layer M3 (lower antenna ratio) Higher metal layers are less sensitive to antenna.

(D) Connect earlier Connect to gate or diffusion earlier in the route.

Key Takeaways:

- Antenna effect is caused by charge buildup on long metal wires during plasma etch.
- Antenna rules in technology data define safe limits (ratios) for each layer.
- Antenna area ratio = metal area / gate area.
- Routing can create violations when long unconnected wires are present.
- Violations are fixed using antenna diodes, smaller segments, higher metal layers, or earlier connections.

Follow the Rules Protect the Device Ensure Reliability!

Cell LEF Fundamentals

Physical view of cells for placement and routing

From Library to Layout!

1 What is Standard Cell LEF ?

- Standard Cell LEF describes the physical information of standard cells (e.g., NAND, NOR, Flip-Flop) used in a standard cell library.
- It provides the layout abstract of the cell for placement and routing.
- It does not contain timing information (that comes from .lib).
- Used for logic cells in the core area.

Input pins (A, B) Height Width Output pin (Y)

Standard Cell (Layout View)

2 What is Macro LEF ?

- Macro LEF describes the physical information of large blocks such as SRAMs, ROMs, PLLs, GPUs, or custom IPs.
- It provides the abstract view, including block size, pin locations, and obstruction (blockages).
- Used for hard macros or soft macros in the design.
- Much larger in size compared to standard cells.

Input pins (address, control) Output pins (data)

SRAM (Macro)

Macro (Top View)

3 What physical information does Cell LEF contain ?

Cell LEF contains the physical abstract needed for P&R tools:

Example (Standard Cell LEF)

```
MACRO NAND2_X1
CLASS CORE ;
ORIGIN 0 0 ;
SIZE 0.46 BY 2.72 ;
SYMMETRY X Y ;
SITE core ;
PIN A
DIRECTION INPUT ;
PORT
LAYER M1 ;
RECT 0.02 1.10 0.06 1.30 ;
END A
...
END NAND2_X1
```

Key Physical Information

- Cell name and type (MACRO, CLASS)
- Size (width, height)
- Pins (name, direction, layer, shape)
- Placement information (site, symmetry, orientation)
- Obstructions / blockages
- Metal layer information (pin shapes)
- Power/Ground pins
- For macros: halo, keepout, blockages, etc.

4 How does Cell LEF connect with the netlist ?

- The netlist instantiates cells by their cell names (e.g., NAND2_X1, DFF_X1).
- The P&R tool uses the LEF file to get the physical view of each cell in the netlist.
- Thus, LEF provides the geometry, and the netlist provides the connectivity.
- Without LEF, the tool would not know the size, pins, and placement details of the cells in the netlist.

Netlist (Verilog)

```
module top (input a, b, output y);
NAND2_X1 U1 (
.A(a),
.B(b),
.Y(y)
);
endmodule
```

Cell name (NAND2_X1) Connects to LEF

Cell LEF (Extract)

```
MACRO NAND2_X1
CLASS CORE ;
SIZE 0.46 BY 2.72 ;
PIN A ... END A
PIN B ... END B
PIN Y ... END Y
END NAND2_X1
```

The cell name in the netlist must match the cell name in the LEF.

5 How does Cell LEF connect with .lib ?

- LEF provides the physical information (geometry, pins, size).
- .lib provides the timing, power, and functional information.
- Both files use the same cell name, so the tools can match the physical and timing views.
- During P&R, the tool reads LEF for placement/routing and uses .lib for timing analysis, optimization, and setup/hold checks.

Cell LEF (Physical View) NAND2_X1

Common Cell Name NAND2_X1

Same cell name connects physical and timing views.

.lib File (Timing View)

```
cell (NAND2_X1) {
area : 1.25;
pin (A) { direction : input; ... }
pin (B) { direction : input; ... }
pin (Y) { direction : output; ... }
timing() { ... }
power() { ... }
}
```

Timing (delay, slew, constraints)
Power (leakage, dynamic)
Functional information

Key Takeaways:

- Standard Cell LEF and Macro LEF provide physical abstracts for P&R.
- Cell LEF contains size, pins, layers, and placement information.
- The netlist uses the same cell names to instantiate cells from the LEF.
- The .lib file provides timing and power information for the same cells.
- Both LEF and .lib are required for a complete and accurate design flow.

Same Cell. Two Views. Complete Design!

Anatomy of a 'MACRO' Block in LEF

Defines the physical abstract view of a hard block (e.g., SRAM, ROM, PLL, IP).

Blocks Build Bigger Chips!

1 What is a 'MACRO' block ?

- A 'MACRO' block in LEF represents a hard IP or large block such as SRAM, ROM, PLL, Analog IP, or custom block.
- It provides the physical abstract view of the block for P&R tools (size, pins, blockages, etc.).
- It does not contain internal details (no transistors or detailed geometry).
- Used for block-level placement, routing around it, and design integration.

Input pins (address, control) Output pins (data)

SRAM (MACRO) Substrate

2 What is 'CLASS' ?

- 'CLASS' defines the type of the macro.
- Common classes: BLOCK (default), CORE, PAD, RING, END CAP, etc.
- Most hard macros like SRAM, ROM, PLL are of class BLOCK.

Example

```
CLASS BLOCK ;
```

Tells the tool that this macro is a hard block.

3 What is 'ORIGIN' ?

- 'ORIGIN' specifies the (X, Y) coordinate of the lower-left corner of the macro.
- It defines the reference point for all geometries (pins, obstructions, etc.) inside the macro.
- Usually (0, 0), but can be non-zero.

Example

```
ORIGIN 0 0 ;
```

Lower-left corner of the macro is at (0, 0).

(0, 0) (ORIGIN)

4 What is 'SIZE' ?

- 'SIZE' defines the width and height of the macro in microns.
- It tells the tool how much area the macro occupies on the chip.
- Used for placement, floorplanning, and routing keep-out.

Example

```
SIZE 100.0 BY 50.0 ;
```

Height =

5 What is 'SYMMETRY' ?

- 'SYMMETRY' defines the allowed orientations of the macro during placement.
- It specifies which rotations and/or mirrors are legal.
- Helps the tool know how the macro can be placed.

Example

```
SYMMETRY X Y ;
```

This macro can be mirrored in X, mirrored in Y, and rotated by 180° (R180).

6 What is 'SITE' ?

- 'SITE' defines the placement grid or row structure used for the macro.
- It indicates the site name (from tech LEF) on which the macro is aligned.
- For standard cells, it is usually a core site (e.g., CORE).
- For macros, it can be a special site (e.g., MACRO, BLOCK, or a site defined in tech LEF).

Example

```
SITE MACRO ;
```

This macro will be placed on the site named 'MACRO'.

Complete Example (Macro LEF)

```
MACRO SRAM_1M
CLASS BLOCK ;
ORIGIN 0 0 ;
SIZE 100.0 BY 50.0 ;
SYMMETRY X Y ;
SITE MACRO ;
PIN A
DIRECTION INPUT ;
PORT
LAYER M1 ;
RECT 0 10 2 20 ;
END A
END SRAM_1M
```


distribution.

- Ensures consistent pin locations (same vertical tracks).
- Defined by the technology (SITE height).

Standard cells with same height (side view)

Same height (e.g., 1.44 μm)

Substrate

All standard cells have the same height, so they can be placed next to each other in a row.

Standard cell row (side view)

Row height = SITE height (e.g., 1.44 μm)

Substrate

Standard cells with varying width (top view)

Width (e.g., 1.0 μm) Width (e.g., 2.4 μm) Width (e.g., 3.6 μm) Width (e.g., 4.8 μm)

Cell width varies based on functionality, but height remains the same for all cells.

4 How do cells fit into placement rows ?

- Cells are placed abutment (edge-to-edge) inside rows.
- They align to the row height and power rail structure.
- The placement tool snaps cells to the row grid defined by the SITE.
- No vertical gaps between cells; only horizontal spacing if needed (e.g., for routing or DRC).

Top view of a placement row

Cells placed abutment in the row (same height, varying width)

5 What happens if cell height mismatches the SITE ?

- If a cell height is different from the SITE (row height):
 - It cannot be placed in the standard row.
 - It may cause placement failure, overlaps, or DRC violations.
 - It breaks the alignment of power rails and routing tracks.
- Such cells are usually treated as special cells (macros) and placed separately.

Height mismatch (problem)

Row height (e.g., 1.44 μm)

Cell height does not match the row height → placement error / DRC violation.

Correct height (okay)

Row height (e.g., 1.44 μm)

All cells match the row height → valid placement.

Key Takeaways:

- Standard cells usually have the same height for abutment, alignment and consistent routing.
- Row height equals the SITE height defined in the LEF.
- Cell width can vary based on functionality.
- Cells fit into placement rows edge-to-edge and align to the row grid.
- Height mismatch causes placement failure and DRC issues.

Aligned Rows Enable Better Performance!

Symmetry and Cell Orientation in LEF

Controls how a cell can be flipped or rotated during placement.

Right Orientation Right Connections Better Chips!

1 What is 'SYMMETRY X' ?

- 'SYMMETRY X' means the cell can be mirrored (flipped) across the X-axis (horizontal flip).
- This creates the MX orientation.
- Useful when rows are placed with alternating orientations.

Mirror across X-axis (MX)

Original (R0) MX (flipped in X)

2 What is 'SYMMETRY Y' ?

- 'SYMMETRY Y' means the cell can be mirrored across the Y-axis (vertical flip).
- This creates the MY orientation.
- Less commonly used for standard cells, but supported in LEF.

Mirror across Y-axis (MY)

Original (R0) MY (flipped in Y)

3 Why are cells flipped in placement rows ?

- Standard cells in adjacent rows are flipped (typically R0 and MX).
- This allows continuous power rails (VDD and VSS) to run in the same horizontal tracks across rows.
- It improves routing efficiency and ensures proper power connectivity.

Two placement rows with flipped cells (side view)

Row 2 (MX) Row 1 (R0)

Power rails aligned across rows

4 What orientations can a standard cell have ?

- Depending on the SYMMETRY statement in LEF, a standard cell can have one or more orientations.
- Common orientations:

Standard cell orientations

R0 (0° rotation) MX (mirror in X) MY (mirror in Y) R180 (180° rotation)

Available orientations depend on the LEF SYMMETRY (e.g., X, Y, or XY).

5 Why is orientation important for power rails ?

- Power rails (VDD/VSS) are typically placed in fixed horizontal tracks in the placement rows.
- Flipping cells (e.g., using MX) ensures that VDD pins align to the VDD rail and VSS pins align to the VSS rail in every row.
- If the orientation is not correct, power pins may end up on the wrong rail, causing:
 - Connectivity errors
 - DRC violations

Correct orientation (alternating R0 / MX)

Row 2 (MX) Row 1 (R0)

VDD and VSS align correctly in all rows.

Wrong orientation (no flip)

Row 2 (MX) Row 1 (R0)

Power rails do not align (VDD/VSS mismatch).

Key Takeaways:

- 'SYMMETRY X' allows horizontal flip (MX).
- 'SYMMETRY Y' allows vertical flip (MY).
- Cells are flipped in rows to keep power rails continuous.
- Standard cells can have R0, MX, MY, and R180 orientations (depending on LEF).
- Correct orientation is essential for proper power rail connection and a functional design.

Same Power Rails Stronger Design Happier Signoff!

Understanding the 'PIN' Block in LEF

Describes the physical interface (pins) of a cell for placement and routing.

Pins Connect Blocks to Build Bigger Chips!

1 What is a LEF PIN ?

- A LEF PIN represents a physical I/O pin of a cell (standard cell or macro).
- It defines where the pin is located on the cell, on which metal layer(s) it exists, and its physical shape (geometry).
- It connects the logical port in the netlist to a physical location for routing.
- Pins are used by the P&R tool (e.g., Innovus) for placement, routing and design rule checking.

NAND2_X1 (Top View)

Pins (as defined in LEF)

2 What information does a PIN block contain ?

- Pin name (e.g., A, Y, VDD)
- Direction (INPUT, OUTPUT, INOUT)
- Use (SIGNAL, POWER, GROUND, etc.)
- Port definition (geometry of the pin)
- Metal layer(s) and shape (RECT, POLYGON, etc.)
- Location and size of the pin
- Additional properties (e.g., antenna, resistance, etc.)

Example PIN block (LEF)

```

PIN A
DIRECTION INPUT ;
USE SIGNAL ;
PORT
  LAYER M1 ;
  RECT 0.10 0.20 0.30 0.40 ;
END
  
```

This defines a pin named 'A', which is an input signal, with a rectangular shape on metal layer M1.

3 What is 'DIRECTION' ?

- DIRECTION specifies the logical direction of the pin.
- Common values: INPUT, OUTPUT, INOUT, FEEDTHRU.
- It tells the tool how the pin is used in the design and helps with connectivity checks.

Example

DIRECTION OUTPUT ;

This pin drives a signal out of the cell.

Direction	Meaning
INPUT	Pin receives a signal
OUTPUT	Pin drives a signal
INOUT	Bidirectional pin
FEEDTHRU	Internal connection (passes through the cell)

4 What is 'USE' ?

- USE specifies the purpose of the pin.
- Common values: SIGNAL, POWER, GROUND, CLOCK, ANALOG, TIEOFF, SCAN, etc.
- It helps the tool understand how the pin should be treated during placement and routing.

5 What is a 'PORT' ?

- PORT defines the physical geometry (shape and layer) of the pin.
- A pin can have one or more PORTS (on different layers).
- Each PORT contains one or more layer shapes (e.g., RECT, POLYGON).

Complete Example and Visualization

```

PIN Y
DIRECTION OUTPUT ;
USE SIGNAL ;
PORT
  LAYER M1 ;
  RECT 1.10 0.80 1.30 1.00 ;
  LAYER M2 ;
  RECT 1.12 0.82 1.28 0.98 ;
END
  
```

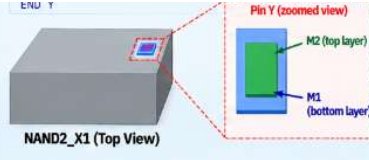
Use	Meaning	Example
SIGNAL	Normal signal pin	
POWER	Power supply pin (e.g., VDD)	USE POWER ;
GROUND	Ground pin (e.g., VSS)	This pin is a power pin (e.g., VDD).
CLOCK	Clock pin	
ANALOG	Analog signal pin	
TIEOFF	Tie high/low cell pin	
SCAN	Scan chain pin	

Example PORT

```

PORT
  LAYER M1 ;
    RECT 0.10 0.20 0.30 0.40 ;
  LAYER M2 ;
    RECT 0.12 0.22 0.28 0.38 ;
END
  
```

This pin has two shapes: one on M1 and one on M2 (used for multi-layer pins).



Key Takeaways:

- A LEF PIN defines the physical interface of a cell.
- A PIN block contains DIRECTION, USE, PORT and geometry information.
- DIRECTION tells how the pin is used (input/output).
- USE defines the purpose of the pin (signal, power, ground, etc.).
- PORT specifies the physical shape and layer(s) of the pin.

Well Defined Pins Enable Correct Connections!

Pin Direction and Pin Use in LEF

Tells the tool how a pin is used and helps in correct placement, routing and timing.

Correct Pin Information Builds Better Chips!

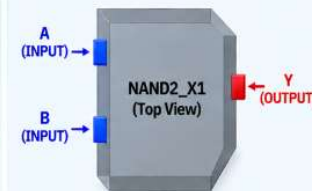
- What is 'INPUT' ?**
 - 'INPUT' means the pin receives a signal into the cell.
 - It is a unidirectional input to the cell.
 - Examples: A, B (inputs of a logic gate), D (data input of a flip-flop).

LEF Example: **DIRECTION INPUT ;**
- What is 'OUTPUT' ?**
 - 'OUTPUT' means the pin drives a signal out of the cell.
 - It is a unidirectional output from the cell.
 - Examples: Y (output of a logic gate), Q (output of a flip-flop).

LEF Example: **DIRECTION OUTPUT ;**
- What is 'INOUT' ?**
 - 'INOUT' means the pin can both receive and drive a signal.
 - It is bidirectional.
 - Examples: I/O pads, bidirectional bus pins (e.g., FPGA/SoC I/O).

LEF Example: **DIRECTION INOUT ;**

Pin Direction in a Logic Cell (Example)



- What is 'USE SIGNAL' ?**
 - 'USE SIGNAL' means the pin is used for normal data signal transfer.
 - It is the most common use type for functional pins.
 - Examples: inputs, outputs, internal signals of logic cells (A, B, Y, D, Q, etc.).

LEF Example: **USE SIGNAL ;**

Indicates that this is a data pin used for functional signal transfer.
- What is 'USE CLOCK' ?**
 - 'USE CLOCK' means the pin is a clock pin.
 - It indicates that this pin carries a clock signal used for sequential elements (flip-flops, latches).
 - Examples: CLK pin of a flip-flop, clock input of a latch.

LEF Example: **USE CLOCK ;**

Identifies this pin as a clock, so the tool can apply special timing and CTS handling.
- Why does CTS care about clock pin information ?**
 - CTS (Clock Tree Synthesis) needs to know which pins are clocks to:
 - Build the clock tree and connect to clock pins.
 - Apply correct clock constraints and buffering.
 - Meet timing, skew and latency requirements.
 - Avoid inserting buffers on non-clock pins.
 - Identify clock gating, generated clocks and special handling (e.g., useful skew, multiple clock domains).

Without correct clock pin information, CTS cannot build a proper clock tree, which can lead to timing failures.

Quick Reference Table

Direction / Use	Meaning	Typical Examples
INPUT	Receives signal into the cell	A, B, D
OUTPUT	Drives signal out of the cell	Y, Q
INOUT	Bidirectional pin	I/O pads, bidirectional bus
USE SIGNAL	Normal data signal pin	A, B, Y, D, Q, etc.
USE CLOCK	Clock signal pin	CLK

Key Takeaways

- Pin direction defines the signal flow (input/output/inout).
- Pin use defines the purpose (signal, clock, etc.).
- Clock pin information is crucial for CTS.
- Correct pin definition in LEF enables proper placement, routing and timing analysis.

Right Pin Information Right Timing A More Reliable Design!

Pin Geometry and Coordinates in LEF

Defines the exact physical location and shape of a pin for routing.

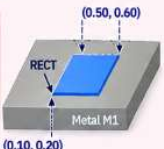
Accurate Pins Enable Correct Routing and a Reliable Design!

- What is 'RECT' ?**
 - 'RECT' defines the rectangular shape of a pin on a specific metal layer.
 - It is specified using two opposite corner coordinates: RECT x1 y1 x2 y2 ;
 - The rectangle represents the physical area where the router can make a connection.

Example (LEF):

```

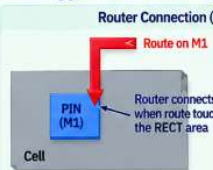
PORT
  LAYER M1 ;
    RECT 0.10 0.20 0.50 0.60 ;
END
  
```


- What do pin coordinates mean ?**
 - Pin coordinates are in microns and are relative to the cell origin (ORIGIN 0 0).
 - (x1, y1) is the lower-left corner and (x2, y2) is the upper-right corner of the rectangle.
 - They define the exact location and size of the pin on the specified metal layer.

Coordinate System (Top View)


- How does the router connect to a pin ?**
 - The router connects to the pin through the defined RECT area on the specified metal layer.
 - Any route that touches (intersects) this rectangle is considered electrically connected to the pin.
 - The router uses these pin shapes to find legal access points during global and detailed routing.

Router Connection (Top View)

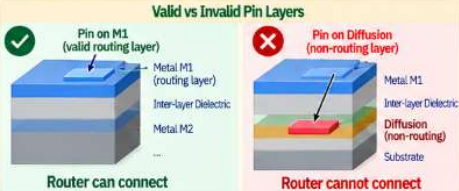


Legend

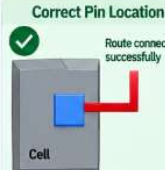
 - Pin shape (RECT) on M1
 - Routed wire on M1
 - Cell boundary

- Why must pins be on routing layers ?**
 - Pins must be defined on valid routing layers (e.g., M1, M2, ...) so that the router can connect to them.
 - These layers are part of the interconnect stack and allow metal wires to be routed.
 - Pins cannot be defined on non-routing layers (e.g., diffusion, implant) because the router cannot place wires there.

Valid vs Invalid Pin Layers


- What happens if a pin location is wrong ?**
 - If the pin is at the wrong location or has an incorrect size:
 - The router may not be able to connect to the pin.
 - It can cause open nets (no connection).
 - It may create design rule violations (DRC).
 - It can lead to functional failure of the chip.

Correct Pin Location



Wrong Pin Location



Key Takeaways

- 'RECT' defines the rectangular pin shape on a routing layer.
- Pin coordinates specify the exact location and size (in microns).
- The router connects when the route touches the pin rectangle.
- Pins must be on valid routing layers.
- Incorrect pin location can cause open nets, DRC errors and functional failure.

Correct Pin Geometry Leads to Smooth Routing and a Working Chip!

VDD and VSS Pins in LEF

Define the power and ground connections of a standard cell for proper placement, routing and power distribution.

- How are VDD and VSS represented in LEF ?**
- What is 'USE POWER' ?**
- What is 'USE GROUND' ?**

Power Pins Power the Design For a Better Tomorrow!

DEFINITION

- VDD and VSS are defined as PINs in the LEF file, just like any other pin.
- They are typically named 'VDD' (power) and 'VSS' (ground).
- Their geometry is specified using RECT on one or more metal layers (usually top metal layer of the cell).
- USE is set to POWER for VDD and GROUND for VSS.

Example LEF (VDD and VSS pins)

PIN VDD

DIRECTION INOUT ;

USE POWER ;

PORT

LAYER M1 ;

RECT 0.00 1.20 2.00 1.40 ;

END VDD

PIN VSS

DIRECTION INOUT ;

USE GROUND ;

PORT

LAYER M1 ;

RECT 0.00 0.00 2.00 0.20 ;

END VSS

Example: VDD Pin

PIN VDD

DIRECTION INOUT ;

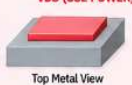
USE POWER ;

PORT

LAYER M1 ;

RECT 0.00 1.20 2.00 1.40 ;

END VDD



Top Metal View (of VDD pin)

Example: VSS Pin

PIN VSS

DIRECTION INOUT ;

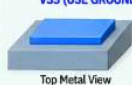
USE GROUND ;

PORT

LAYER M1 ;

RECT 0.00 0.00 2.00 0.20 ;

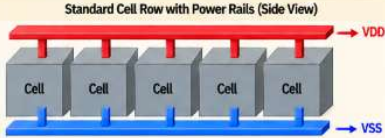
END VSS



Top Metal View (of VSS pin)

4 Why are power rails important in standard cells ?

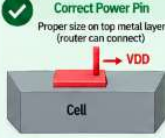
- Power rails (VDD and VSS) supply the required voltage and ground to all the transistors inside the cell.
- They run horizontally across placement rows and connect to the power pins of each cell.
- Ensure reliable power delivery, low resistance and low IR drop.
- Without proper power rails, the cell cannot function.



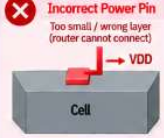
Standard Cell Row with Power Rails (Side View)

5 What happens if power pin geometry is incorrect ?

- If the pin is missing, too small or not on the correct metal layer, the power rail may not connect properly.
- This can cause:
 - Open power/ground (no connection)
 - High resistance (IR drop)
 - DRC violations
 - Functional failure (cell may not work)
 - Signal integrity and reliability issues

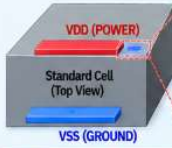


Correct Power Pin
Proper size on top metal layer (router can connect)



Incorrect Power Pin
Too small / wrong layer (router cannot connect)

Complete View: Standard Cell with VDD and VSS Pins



Standard Cell (Top View)

VDD (POWER)

VSS (GROUND)

Zoomed Pin Geometry (Top View)

VDD on M1
RECT x1 x2 y1 y2

VSS on M1
RECT x1 y1 x2 y2

Typical Layers Used

- Power pins are usually on the top metal layer of the cell (e.g., M1 or higher).
- This allows continuous power rails to run across the row.
- Multiple RECTs or multiple layers can be used for larger power pins.

Key Takeaways:

- VDD and VSS are defined as PINs in LEF with USE POWER and USE GROUND respectively.
- Power rails connect to these pins to supply voltage and ground.
- Correct pin geometry (size and layer) is essential for proper connection.
- Incorrect power pin geometry can lead to open nets, high IR drop, DRC errors and functional failure.

Strong Power Pins Make a Stronger Chip !

Routing Obstruction (OBS) in LEF

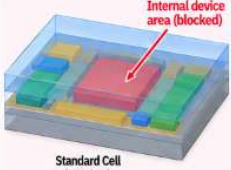
Defines areas where routing is NOT allowed inside a cell.

1 What is 'OBS' ?

- 'OBS' (Obstruction) defines areas inside a cell where routing is not allowed.
- It is specified in the LEF file using rectangular or polygon shapes on specific metal layers.
- These areas are physical blockages (e.g., due to cell structures, devices, or keep-out regions).

Example (LEF)

```
OBS
  LAYER M2 ;
  RECT 0.20 0.50 1.00 1.20 ;
END
END OBS
```



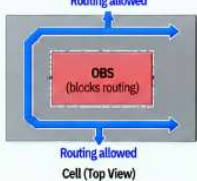
Standard Cell (3D View)

2 Why does OBS exist ?

- Standard cells contain internal structures (transistors, well ties, implants, etc.) that should not have signal wires over them.
- OBS protects these sensitive regions from routing.
- It ensures routability only in allowed (open) areas, maintaining design rule compliance and reliability.

3 What does OBS block ?

- OBS blocks signal routing (wires and vias) on the specified metal layer(s).
- It does not block cell placement (the obstruction is inside the cell).
- It can block one or more layers (e.g., M1, M2, ...).
- Typically defined using RECT or POLYGON.



Routing allowed

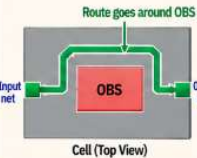
OBS (blocks routing)

Routing allowed

Cell (Top View)

4 How does routing use OBS ?

- During global and detailed routing, the router reads the OBS information from the LEF.
- It treats OBS regions as non-routable areas.
- The router automatically avoids these regions while finding routes.
- Only the open (non-OBS) areas are used for routing.



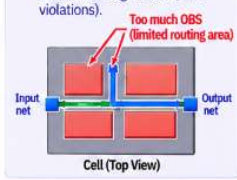
Route goes around OBS

OBS

Cell (Top View)

5 How can too much OBS cause congestion ?

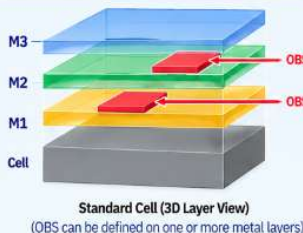
- If a cell has large or excessive OBS areas, the available routing area becomes very limited.
- This forces more wires into fewer tracks, increasing routing congestion.
- It can lead to longer routes, more vias, higher resistance, and timing degradation.
- In extreme cases, the router may fail to find a legal route (DRC violations).



Too much OBS (limited routing area)

Cell (Top View)

Layer View of OBS (Multiple Layers)



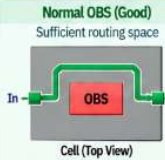
Standard Cell (3D Layer View)

(OBS can be defined on one or more metal layers)

Routing Comparison

Normal OBS (Good)

Sufficient routing space

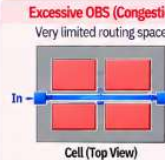


Cell (Top View)

Routing completes easily

Excessive OBS (Congestion)

Very limited routing space



Cell (Top View)

Causes congestion, longer routes or routing failure

Key Takeaways

- OBS defines non-routable regions inside a cell.
- It exists to protect internal structures.
- OBS blocks signal routing on specified metal layers.
- The router avoids OBS and uses only open areas.
- Excessive OBS can cause routing congestion and timing issues.

Right Obstructions Enable Smooth Connections and a Reliable Chip !

Physical Design Builds the Future

Standard Cell vs Macro LEF

Two different types of LEF. Two different roles in physical design.

1 What is the difference between Standard Cell LEF and Macro LEF ?

- Standard Cell LEF describes small, regularly shaped cells used in the core (e.g., gates, flip-flops).
- Macro LEF describes large, pre-designed blocks with fixed size and complex internal structure (e.g., SRAM, PLL, IPs).
- Standard cells are placed in rows and can be flipped (various orientations).
- Macros are placed as fixed or limited-orientation blocks.

Feature	Standard Cell LEF	Macro LEF
Size	Small (µm range)	Large (tens to hundreds of µm)
Placement	In rows, abutted	Placed as a block (usually fixed)
Orientation	Multiple (RD, MX, MY, R180)	Usually fixed (often RD only)
Pins	Usually small, on edges	Can be many pins, on multiple sides/layers
Obstructions	Minimal (within cell)	Detailed OBS (internal + keep-out regions)
Examples	NAND, NOR, INV, DFF, MUX	SRAM, PLL, ADC, CPU, USB IP

2 Why are SRAM and PLL treated as macros ?

- SRAM and PLL are large, complex blocks with internal circuitry (memory arrays, analog circuits, etc.).
- They are usually designed and verified separately (often by IP vendors).
- They have fixed size, specific pin locations, and strict design rules.
- They are not built from standard cells, so they cannot be placed in rows.
- They require special handling for power, clock, and isolation.

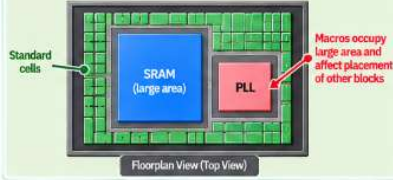


SRAM: Large memory array, pre-designed IP

PLL: Complex analog/mixed-signal block, pre-designed IP

3 How does macro size affect floorplanning ?

- Macros occupy a large area, so they must be placed early during floorplanning.
- Their size determines the available space for standard cells, routing channels, and other macros.
- Large macros can create blockages and may require more whitespace around them (keep-out region).
- Proper placement helps optimize wire length, timing, and power distribution.



Floorplan View (Top View)

Standard cells

SRAM (large area)

PLL

Macros occupy large area and affect placement of other blocks

4 How do macro pin locations affect routing ?

- Macro pins are usually placed on specific sides and metal layers.
- Routing has to reach these fixed pin locations, which can create

5 Why does a macro require obstruction information ?

- Macros contain internal structures (memory arrays, analog blocks, well ties, etc.) that should not be routed over.

Example LEF Snippets

Standard Cell LEF (simplified)

```
DEFINITION
  LAYER M1 ;
  RECT 0.00 0.00 1.00 1.00 ;
END
```

Macro LEF (simplified)

```
DEFINITION
  LAYER M1 ;
  RECT 0.00 0.00 10.00 10.00 ;
END
```

Different Blocks, Different Purposes, A Complete Chip !

Physical Design Builds the Future

congestion if many pins are in a small area.

- Pin locations influence macro placement (e.g., place macro closer to blocks it connects to).
- Poor pin placement can lead to longer routes, more vias, higher resistance, and timing violations.

Good Pin Placement

- Shorter routes
- Less congestion
- Better timing
- Easier power distribution

Bad Pin Placement

- Longer routes
- Routing congestion
- Timing issues
- More vias and resistance

Macro Pin Locations (Top View)

Macro with Obstruction (Top View)

```

MACRO NAND2_X1
CLASS CORE ;
ORIGIN 0 0 ;
SIZE 1.2 BY 2.0 ;
SYMMETRY X Y ;
PIN A
DIRECTION INPUT ;
USE SIGNAL ;
PORT
LAYER M1 ;
RECT 0.1 0.2 0.2 0.6 ;
END A
END NAND2_X1

MACRO SRAM_256X32
CLASS BLOCK ;
ORIGIN 0 0 ;
SIZE 100.0 BY 80.0 ;
SYMMETRY X Y ;
PIN A[8]
DIRECTION INPUT ;
USE SIGNAL ;
PORT
LAYER M4 ;
RECT 0.0 10.0 5.0 12.0 ;
END A[8]
OBS
LAYER M1 ;
RECT 0.0 0.0 100.0 80.0 ;
END SRAM_256X32

```

Key Takeaways

- Standard cells are small, placed in rows, and used for core logic.
- Macros are large, pre-designed blocks with fixed size and special requirements.
- Macro size, pin locations, and OBS information directly impact floorplanning, routing, timing, and overall chip success.
- Proper use of LEF information ensures a reliable, manufacturable, and high-performance design.

Plan Macros Early, Route Smartly, Build a Better Chip!

Macro Placement and Halo in LEF

LEF provides the necessary information to place macros correctly and keep enough space around them for routing.

1 What information does LEF provide for macro placement?

- Macro LEF provides the physical size (WIDTH, HEIGHT).
- The ORIGIN (reference point) for placement.
- The CLASS (e.g., BLOCK).
- The SYMMETRY (allowed orientations).
- The MACRO PIN locations (for routing).
- OBS (obstructions) and optionally placement blockages inside the macro.

Example Macro LEF (Simplified)

```

MACRO SRAM_256X32
CLASS BLOCK ;
ORIGIN 0 0 ;
SIZE 100.0 BY 80.0 ;
SYMMETRY X Y ;
PIN A[8]
DIRECTION INPUT ;
PORT
LAYER M4 ;
RECT 0.0 10.0 5.0 12.0 ;
END A[8]
...
END SRAM_256X32

```

Macro with size from LEF (3D view)

2 What is macro halo?

- A macro halo (keep-out margin) is an extra spacing around the macro where standard cells and/or routing are not allowed.
- It is defined by the designer (not usually in LEF, but in floorplanning/placement tools using LEF as reference).
- It gives space for routing, power straps, and design rule compliance.

Macro Halo (Top View)

3 What is placement blockage?

- Placement blockage is a region where standard cells cannot be placed.
- It can be defined manually by the designer (e.g., around macros, analog blocks, or for future use).
- It can be larger than the macro halo and is used only for placement (not for routing, unless specified).

Placement Blockage around a Macro (Top View)

4 How do OBS and halo differ?

Parameter	OBS (Obstruction)	Halo
Purpose	Blocks routing (and sometimes placement) inside the macro or specific regions.	Keeps space around the macro (mainly for placement, sometimes for routing).
Defined In	In LEF (inside MACRO definition).	Defined in floorplanning/placement tool (not in LEF).
Blocks	Routing (and optionally placement).	Placement of standard cells (and sometimes routing).
Location	Inside the macro or specific internal regions.	Outside the macro (around it).
Usage	Represents real physical blockages (e.g., internal circuitry, keep-out regions).	Provides spacing for routing, power, and design rule compliance.
Example	OBS LAYER M2 ; RECT 10.0 10.0 30.0 30.0 ; END	setHaloToBlock -type hard -size (10 10 10) [get_cell SRAM_256X32]

5 Why are macros important for congestion?

- Macros are large blocks that occupy significant area and reduce the available space for standard cells and routing.
- They create routing bottlenecks, especially if many pins are on one side.
- Improper macro placement can lead to high congestion, longer routes, and timing violations.
- Good macro placement, with proper halo and blockages, helps distribute routing resources and reduces congestion.

Poor Macro Placement (High Congestion) vs Good Macro Placement (Less Congestion)

Complete Floorplan View

Chip Floorplan (Top View)

Key Takeaways

- LEF provides macro size, origin, pins, and OBS for correct placement.
- Macro halo keeps space around the macro for routing and design rules.
- Placement blockage prevents standard cells from being placed in a region.
- OBS blocks routing (and sometimes placement) inside the macro, halo is outside the macro.
- Macros significantly affect congestion; proper placement and spacing improve routability, timing, and overall chip performance.

Plan Smart, Route Easy!

How Innovus Reads LEF

From Technology to Cells – Building the Design Database

1 How does Innovus read Tech LEF?

- Innovus reads Tech LEF using the command `ldefin <tech_lef_file>`.
- It parses the technology information such as layers, layer types (ROUTING, CUT, MASTERSLICE), layer properties (width, spacing, pitch, direction), via definitions, and manufacturing grid.
- This information is stored in Innovus technology database and is used for DRC rules, routing, and physical verification during P&R.

Example Command

```
ldefin /path/to/tech.lef
```

2 How does Innovus read Cell LEF?

- Innovus reads Cell LEF using the same command `ldefin <cell_lef_file>`.
- It parses information about each cell such as size (WIDTH, HEIGHT), pins (PIN, DIRECTION, USE, PORT, layer geometry), obstructions (OBS), symmetry, and site information.
- The cell LEF information is stored in the design database and used for placement, routing, and timing analysis.

Example Command

```
ldefin /path/to/cells.lef
```

3 Why must Tech LEF be loaded first?

- Cell LEF references technology layers defined in Tech LEF (e.g., M1, M2, VIA1, etc.).
- If Tech LEF is not loaded first, Innovus does not know what these layers are.
- Tech LEF also defines units, manufacturing grid, and other tech rules that are required to interpret the cell geometries correctly.
- Therefore, Tech LEF must be loaded before any Cell LEF.

Correct Order

- ldefin tech.lef # Load technology information
- ldefin stdcell.lef # Load standard cell library

Cell LEF (uses layers from Tech LEF)

Tech LEF (defines layers, rules)

Innovus Database

4 What happens if Cell LEF references an unknown layer?

- If a Cell LEF uses a layer that is not defined in the Tech LEF, Innovus will report an error and fail to read that cell LEF.
- The cell will not be loaded into the design database.
- You must load the correct Tech LEF that contains all the layers used by the cell LEF (or use a compatible version).

Example Error Message

```

**ERROR: (LEF-004) Layer 'M9' referenced in cell LEF 'stdcell.lef' is not defined.
Please load the technology LEF with this layer.

```

Fix

- Load the correct Tech LEF.
- Check for layer name mismatches (e.g., M9 vs METAL9).

5 How do you check whether LEF loaded correctly?

- Innovus provides several commands to verify the loaded LEF information:

Command	What it Shows
reportTechnology	Shows technology information (layers, vias, etc.)
reportLayers	Lists all layers and their properties
reportLibs	Lists all loaded cell libraries
reportCells	Shows all cells read from LEF
reportCell <cell_name>	Detailed information about a specific cell (size, pins, OBS, etc.)
dbGet top.fplan.layers.name	Tcl command to list layer names
dbGet top.lib.cells.name	Tcl command to list loaded cells

Example Outputs (Simplified)

```

Innovus> reportLayers
Layer Name Type Direction
M1 ROUTING HORIZONTAL
M2 ROUTING VERTICAL
M3 ROUTING HORIZONTAL

Innovus> reportCells
Number of cells: 842
First 8 cells:
NAND2_X1
NAND2_X1
OFF_X1
AOE21_X1

```

Key Takeaways

- Tech LEF defines the technology (layers, vias, rules) and must be loaded first.
- Cell LEF defines the cells (size, pins, OBS, etc.) and is loaded next.
- Unknown layers in Cell LEF will cause an error.
- Use report commands to verify that LEF is loaded correctly.
- Correct LEF loading ensures successful placement, routing and signoff.

Correct LEF, Smoother P&R, Better Chips!

LEF in Floorplanning and Power Planning

LEF provides the physical and electrical information needed to plan the chip layout and power distribution.

1 How does LEF help calculate physical area?

- LEF provides the size (WIDTH, HEIGHT) of each cell and macro.
- It also gives the site definition and row height.
- Using this information, the tool calculates the exact physical area required for cells and macros.

Example (Standard Cell)

```
MACRO INV_X1
CLASS CORE ;
SIZE 1.20 BY 2.00 ;
SYMMETRY X Y ;
...
END INV_X1
```

Physical area = $1.20 \times 2.00 = 2.4 \mu\text{m}^2$

2 How does LEF help macro placement?

- LEF defines macro size, shape, and placement blockages (if any).
- It also provides pin locations and orientation rules.
- The tool uses this information to place macros in the floorplan while respecting space, blockages, and design constraints.

Floorplan with macros placed using LEF information

3 How does SITE create standard cell rows?

- SITE defines the standard cell height, width, and routing tracks.
- It tells the tool how to create rows and align cells on the placement grid.
- This ensures proper power connections, routing access, and cell alignment.

Example SITE definition

```
SITE CORE
CLASS CORE ;
SIZE 0.20 BY 0.50 ;
SYMMETRY Y ;
END CORE
```

SITE → Creates aligned standard cell rows

4 How does LEF help create power rails?

- LEF provides the geometry and placement of VDD/VSS pins and power stripes for macros and standard cells.
- The tool uses this information to create continuous power rails and connect the power grid.
- It ensures proper spacing, metal layers, and alignment for reliable power distribution.

Power rails created using LEF information

5 How do VDD/VSS pin locations affect power connectivity?

- The exact location of VDD/VSS pins (from LEF) determines where the power rails must be placed.
- Correct pin locations ensure easy and short connections to the power grid.
- If pin locations are wrong, it can lead to power shorts, open connections, or extra routing and area overhead.

Correct VDD/VSS pin locations

- Good Connectivity
- Short connections
- Clean power grid
- Reliable operation

Wrong VDD/VSS pin locations

- Poor Connectivity
- Longer routes
- Possible shorts
- Higher resistance

Key Takeaways

- LEF provides size, shape, pin, and power information useful for floorplanning and power planning.
- Accurate LEF data ensures proper area calculation, macro placement, standard cell rows, and power rails.
- Correct VDD/VSS pin locations are critical for reliable power connectivity and performance.

LEF in Placement and CTS

From library information to a high-performance, timing-clean design

1 How does LEF ensure legal placement?

- LEF defines the SIZE (width, height) of each cell.
- LEF specifies the SITE, which sets the row height, row site width, and legal placement grid.
- Each cell has a site name, so it can only be placed on rows of the same site.
- The tool uses this information to snap cells to the placement grid, ensuring legal placement (no overlap, correct orientation).

LEF (Cell)

```
MACRO INV_X1
CLASS CORE ;
SIZE 0.46 BY 1.20 ;
SITE CoreSite ;
SYMMETRY X Y ;
END INV_X1
```

Legal placement (aligned to site grid)

2 How do pin locations help timing-driven placement?

- LEF provides the exact pin locations (layer and coordinates) for each cell.
- The placer uses pin locations to estimate interconnect (wire) length between cells.
- Cells are placed closer if they have critical timing connections to reduce wire delay.
- Accurate pin locations help the tool make better timing-driven placement decisions.

Closer placement → shorter wire → better timing

3 How does OBS affect placement congestion?

- OBS (obstruction) defines areas inside a cell or macro where standard cells cannot be placed.
- The placer treats OBS regions as blocked areas.
- Large or poorly defined OBS regions reduce the available placement area, which can increase placement congestion.
- This may lead to higher cell density in other areas, longer wire lengths, and potential timing issues.

Normal Placement (Less OBS)

More space for placement → Lower congestion

Large OBS (Higher Congestion)

Less space for placement → Higher congestion

4 How does CTS identify clock pins?

- LEF defines the USE CLOCK attribute for clock pins.
- During CTS, the tool reads this information to identify which pins are clock pins.
- These pins are used as sources or sinks for the clock tree depending on their direction (input/output).
- Without correct clock pin information, CTS may not detect clock pins, leading to an incomplete or incorrect clock tree.

LEF (Clock Pin Example)

```
PIN CLK
DIRECTION : INPUT ;
USE : CLOCK ;
PORT
LAYER M4 :
RECT 0.0 0.0 0.2 0.2 ;
END
END CLK
```

Clock Tree built using clock pin information

5 How do physical layer rules affect clock routing?

- LEF (along with Tech LEF) defines the available layers, directions, widths, spacing, and other physical rules.
- CTS uses these rules to route the clock with proper width, spacing, and layer selection.
- Clock nets usually require wider wires and higher metal layers to reduce RC delay and ensure low skew.
- The tool must follow design rules (from Tech LEF) to avoid DRC violations during clock routing.
- If a layer is unidirectional (e.g., M1 horizontal), CTS will use allowed directions to build the clock tree.

Tech LEF (Layer Definition)

```
LAYER M4
TYPE ROUTING ;
DIRECTION HORIZONTAL ;
PITCH 0.80 ;
WIDTH 0.40 ;
SPACING 0.40 ;
END M4
```

Clock Routing Example (Top View)

Key Takeaways

- LEF ensures legal placement using cell size and site information.
- Accurate pin locations help timing-driven placement.
- OBS reduces available placement area and can increase congestion.
- CTS identifies clock pins using the USE CLOCK attribute from LEF.
- Physical layer rules (from Tech LEF) guide clock routing with proper width, spacing, and layer usage.

First Checks After Receiving LEF

Basic sanity checks to ensure the LEF is correct and ready for Physical Design

1 Check file availability?

- Make sure all required LEF files (tech LEF, cell LEF, macro LEF) are received and accessible.
- Check file size and timestamp to confirm they are not empty or corrupted.

Innovus Command (Linux shell)

```
ls -lh *.lef
```

2 Check LEF version?

- Verify the LEF version is supported by Innovus (e.g., 5.6, 5.7, 5.8).
- Mismatched version may cause read errors.

Innovus Command

```
grep -i "VERSION" *.lef
```

3 Check units?

- Verify the database units (e.g., MICRONS 1000 means 1 unit = 0.001 μm = 1 nm).
- Ensure it matches with .lib and other design files.

UNITS

```
DATABASE MICRONS 1000 ;
END UNITS
```

Innovus Command

```
grep -i "UNITS" *.lef
```

4 Check manufacturing grid?

- Check the manufacturing grid (MANUFACTURINGGRID) which defines the legal coordinate grid.
- Ensure it matches with .lib and other design files.
- All shapes must be on this grid for DRC clean design.

Innovus Command

```
grep -i "MANUFACTURINGGRID" *.lef
```

5 Check metal layers?

- Verify available metal layers, their direction (H/V), and basic properties.
- Ensure required routing layers (e.g., M1-M10) are defined.

Innovus Command

```
reportLayers
```

Check Early
Avoid Later Issues
Save Time!

6 Check SITE?

- Verify SITE definition (row height, width, symmetry).
- Ensure it matches the standard cell library and .lib file.

Site height (e.g., 0.48 μm)
Site width (e.g., 0.24 μm)

Innovus Command
reportSite

```
[innovus] reportSite
Site Width Height Symmetry
CoreSite 0.240 0.480 X Y
```

7 Check required macros?

- Ensure all required macros (e.g., SRAM, PLL, IO) are present in the LEF.
- Check their size and pin information.

Innovus Command
reportMacros

```
[innovus] reportMacros
Macro Width Height Class
SRAM_1 128.0 88.0 BLOCK
PLL_0 60.0 60.0 BLOCK
IO_PAD 58.0 200.0 BLOCK
```

8 Check pins?

- Verify pin names, direction (INPUT/OUTPUT/INOUT), use (SIGNAL/CLOCK/POWER/GROUND), layer and coordinates.
- Check for important pins like VDD, VSS, CLK, RESET, etc.

PIN CLK DIRECTION INPUT ;
USE SIGNAL ;
PORT ;
LAYER M3 ;
RECT 10.0 20.0 30.0 20.2 ;
END

Innovus Command
reportPins

```
[innovus] reportPins -n 10
Pin Dir Use Layer Location
CLK IN CLOCK M3 (10.0,20.0)
RST IN SIGNAL M2 (5.0,10.0)
```

9 Check OBS?

- Ensure obstruction (OBS) information is present for cells and macros.
- OBS prevents routing/placement in blocked regions.

Innovus Command
reportObstructions

```
[innovus] reportObstructions -n 5
Type Layer Blob
OBS M1 (0.1 0.1 1.0 0.3)
OBS M2 (0.0 1.2 1.0 1.5)
```

10 Check consistency with netlist and .lib?

- Ensure the cell names in LEF match the netlist and .lib files.
- Check for missing cells or extra cells.

Netlist (.v) ↔ LEF (.lef) ↔ Library (.lib)

Innovus Command
checkDesign -unresolved reportDesignLibs

```
[innovus] checkDesign -unresolved
No unresolved references found.
[innovus] reportDesignLibs
Number of loaded libraries : 3
Number of loaded cells : 842
```

Key Takeaways

- Verify all LEF files are available and accessible.
- Check LEF version and units.
- Confirm manufacturing grid and metal layers.
- Ensure SITE and required macros are defined.
- Validate pins, OBS, and consistency with netlist/.lib.

Good LEF = Smooth PD Flow = Less Debug Time!

Linux Commands for LEF Investigation

Quick and useful commands to explore LEF files in a Physical Design flow

1 How do you list LEF files?

- Use `ls` command to list LEF files in the current directory.
- You can use wildcards (*.lef) to filter only LEF files.

Command
`ls -l *.lef`

Example Output

```
[user@server]$ ls -l *.lef
-rw-r--r-- 1 user user 1.2M Aug 10 10:22 tech.lef
-rw-r--r-- 1 user user 3.8M Aug 10 10:22 stdcell.lef
-rw-r--r-- 1 user user 5.1M Aug 10 10:22 macro.lef
```

2 How do you check LEF file size?

- Use `ls -lh` to see file size in human readable format.
- Useful to check if the file is complete or corrupted.

Command
`ls -lh <file>.lef`

Example Output

```
[user@server]$ ls -lh stdcell.lef
-rw-r--r-- 1 user user 3.8M Aug 10 10:22 stdcell.lef
```

Very small file size may indicate an incomplete or corrupted LEF.

3 How do you find all MACRO definitions?

- Use `grep` to search for MACRO keyword.
- `-n` shows line numbers.

Command
`grep -n "MACRO" <file>.lef`

Example Output

```
[user@server]$ grep -n "MACRO" macro.lef
10:MACRO SRAM_1R1W
256:MACRO PLL_TOP
420:MACRO IO_PAD
689:MACRO ADC
1021:MACRO BTAS_GEN
```

Lists all macro names defined in the LEF.

4 How do you find a specific cell?

- Use `grep` to search for a cell name (MACRO or standard cell).
- Useful when you know the exact cell name.

Command
`grep -n "NAND2_X1" <file>.lef`

Example Output

```
[user@server]$ grep -n "NAND2_X1" stdcell.lef
1023:MACRO NAND2_X1
1024:CLASS CORE ;
1025:SIZE 1.20 BY 2.00 ;
...
1050:END NAND2_X1
```

Finds the full definition of the cell.

5 How do you check cell SIZE?

- Search for the cell name and look for `SIZE` statement.
- You can print a few lines around the match using `-A` (after) or `-B` (before).

`grep -A 5 "NAND2_X1" stdcell.lef`

Example Output

```
[user@server]$ grep -A 5 "NAND2_X1" stdcell.lef
MACRO NAND2_X1
CLASS CORE ;
SIZE 1.20 BY 2.00 ;
SYMMETRY X Y ;
SITE CoreSite ;
...
```

SIZE from LEF

6 How do you find PIN definitions?

- Search for PIN keyword inside a cell.
- Use `-n` for line numbers.

`grep -n "PIN" stdcell.lef`

Example Output

```
[user@server]$ grep -n "PIN" stdcell.lef
1030:PIN A
1040:PIN B
1050:PIN Y
...
```

Example Pin Definition (snippet)

```
PIN A
DIRECTION INPUT ;
USE SIGNAL ;
PORT ;
LAYER M3 ;
RECT 0.0 0.1 0.2 0.3 ;
END
END A
```

7 How do you find OBS?

- Search for OBS keyword.
- Useful to check obstruction geometry inside cells/macros.

`grep -n "OBS" <file>.lef`

Example Output

```
[user@server]$ grep -n "OBS" macro.lef
320:OBS
321:LAYER M4 ;
322:RECT 10.0 20.0 30.0 40.0 ;
323:END
450:OBS
451:LAYER M5 ;
452:RECT 5.0 5.0 15.0 25.0 ;
453:END
```

OBS regions block routing and/or placement.

8 How do you compare .lib and .lef cell names?

- Extract cell names from LEF (MACRO) and .lib, then compare.
- Use `sort` and `comm` to find mismatches.

Get cell names from LEF
`grep "MACRO" stdcell.lef | awk '{print $2}' | sort > lef_cells.txt`

Get cell names from .lib
`grep "cell" libcell1.lib | awk '{print $2}' | sed 's/"/' | sort > lib_cells.txt`

Compare the two lists
`comm -3 lib_cells.txt lef_cells.txt`

Output shows:

- cells only in .lib
- cells only in .lef
- helps find missing or extra cells

Key Takeaways

- Use basic Linux commands (`ls`, `grep`, `awk`, `sort`, `comm`) to quickly analyze LEF files.
- Check file size, version, cells, pins, OBS, and compare with .lib.
- These commands help you catch issues early and save debug time in physical design.

Small Commands Big Debugging Power!

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- Useful to check if the file is complete or corrupted.

Command
`ls -lh <file>.lef`

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Very small file size may indicate an incomplete or corrupted LEF.

3 How do you find all MACRO definitions?

- Use `grep` to search for MACRO keyword.
- `-n` shows line numbers.

Command
`grep -n "MACRO" <file>.lef`

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```
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Lists all macro names defined in the LEF.

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Finds the full definition of the cell.

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- Search for the cell name and look for `SIZE` statement.
- You can print a few lines around the match using `-A` (after) or `-B` (before).

`grep -A 5 "NAND2_X1" stdcell.lef`

Example Output

```
[user@server]$ grep -A 5 "NAND2_X1" stdcell.lef
MACRO NAND2_X1
CLASS CORE ;
SIZE 1.20 BY 2.00 ;
SYMMETRY X Y ;
SITE CoreSite ;
...
```

SIZE from LEF

6 How do you find PIN definitions?

- Search for PIN keyword inside a cell.
- Use `-n` for line numbers.

`grep -n "PIN" stdcell.lef`

Example Output

```
[user@server]$ grep -n "PIN" stdcell.lef
1030:PIN A
1040:PIN B
1050:PIN Y
...
```

Example Pin Definition (snippet)

```
PIN A
DIRECTION INPUT ;
USE SIGNAL ;
PORT ;
LAYER M3 ;
RECT 0.0 0.1 0.2 0.3 ;
END
END A
```

7 How do you find OBS?

- Search for OBS keyword.
- Useful to check obstruction geometry inside cells/macros.

`grep -n "OBS" <file>.lef`

Example Output

```
[user@server]$ grep -n "OBS" macro.lef
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OBS regions block routing and/or placement.

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Compare the two lists
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Output shows:

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- cells only in .lef
- helps find missing or extra cells

SIZE 1.20 BY 2.00 ;
SYMMETRY X Y ;
SITE CoreSite ;
...
1.20 µm (Width)

SIZE FROM LEF

END A
END B
END C
END D
END E
END F
END G
END H
END I
END J
END K
END L
END M
END N
END O
END P
END Q
END R
END S
END T
END U
END V
END W
END X
END Y
END Z
END AA
END AB
END AC
END AD
END AE
END AF
END AG
END AH
END AI
END AJ
END AK
END AL
END AM
END AN
END AO
END AP
END AQ
END AR
END AS
END AT
END AU
END AV
END AW
END AX
END AY
END AZ
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END BB
END BC
END BD
END BE
END BF
END BG
END BH
END BI
END BJ
END BK
END BL
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END BN
END BO
END BP
END BQ
END BR
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END BU
END BV
END BW
END BX
END BY
END BZ
END CA
END CB
END CC
END CD
END CE
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END FJ
END FK
END FL
END FM
END FN
END FO
END FP
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END FS
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END IX
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END JA
END JB
END JC
END JD
END JE
END JF
END JG
END JH
END JI
END JJ
END JK
END JL
END JM
END JN
END JO
END JP
END JQ
END JR
END JS
END JT
END JU
END JV
END JW
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END KQ
END KR
END KS
END KT
END KU
END KV
END KW
END KX
END KY
END KZ
END LA
END LB
END LC
END LD
END LE
END LF
END LG
END LH
END LI
END LJ
END LK
END LL
END LM
END LN
END LO
END LP
END LQ
END LR
END LS
END LT
END LU
END LV
END LW
END LX
END LY
END LZ
END MA
END MB
END MC
END MD
END ME
END MF
END MG
END MH
END MI
END MJ
END MK
END ML
END MN
END MO
END MP
END MQ
END MR
END MS
END MT
END MU
END MV
END MW
END MX
END MY
END MZ
END NA
END NB
END NC
END ND
END NE
END NF
END NG
END NH
END NI
END NJ
END NK
END NL
END NM
END NN
END NO
END NP
END NQ
END NR
END NS
END NT
END NU
END NV
END NW
END NX
END NY
END NZ
END OA
END OB
END OC
END OD
END OE
END OF
END OG
END OH
END OI
END OJ
END OK
END OL
END OM
END ON
END OO
END OP
END OQ
END OR
END OS
END OT
END OU
END OV
END OW
END OX
END OY
END OZ
END PA
END PB
END PC
END PD
END PE
END PF
END PG
END PH
END PI
END PJ
END PK
END PL
END PM
END PN
END PO
END PP
END PQ
END PR
END PS
END PT
END PU
END PV
END PW
END PX
END PY
END PZ
END QA
END QB
END QC
END QD
END QE
END QF
END QG
END QH
END QI
END QJ
END QK
END QL
END QM
END QN
END QO
END QP
END QQ
END QR
END QS
END QT
END QU
END QV
END QW
END QX
END QY
END QZ
END RA
END RB
END RC
END RD
END RE
END RF
END RG
END RH
END RI
END RJ
END RK
END RL
END RM
END RN
END RO
END RP
END RQ
END RR
END RS
END RT
END RU
END RV
END RW
END RX
END RY
END RZ
END SA
END SB
END SC
END SD
END SE
END SF
END SG
END SH
END SI
END SJ
END SK
END SL
END SM
END SN
END SO
END SP
END SQ
END SR
END SS
END ST
END SU
END SV
END SW
END SX
END SY
END SZ
END TA
END TB
END TC
END TD
END TE
END TF
END TG
END TH
END TI
END TJ
END TK
END TL
END TM
END TN
END TO
END TP
END TQ
END TR
END TS
END TT
END TU
END TV
END TW
END TX
END TY
END TZ
END UA
END UB
END UC
END UD
END UE
END UF
END UG
END UH
END UI
END UJ
END UK
END UL
END UM
END UN
END UO
END UP
END UQ
END UR
END US
END UT
END UY
END UZ
END VA
END VB
END VC
END VD
END VE
END VF
END VG
END VH
END VI
END VJ
END VK
END VL
END VM
END VN
END VO
END VP
END VQ
END VR
END VS
END VT
END VY
END VZ
END WA
END WB
END WC
END WD
END WE
END WF
END WG
END WH
END WI
END WJ
END WK
END WL
END WM
END WN
END WO
END WP
END WQ
END WR
END WS
END WT
END WY
END WZ
END XA
END XB
END XC
END XD
END XE
END XF
END XG
END XH
END XI
END XJ
END XK
END XL
END XM
END XN
END XO
END XP
END XQ
END XR
END XS
END XT
END XU
END XV
END XW
END XX
END XY
END XZ
END YA
END YB
END YC
END YD
END YE
END YF
END YG
END YH
END YI
END YJ
END YK
END YL
END YM
END YN
END YO
END YP
END YQ
END YR
END YS
END YT
END YU
END YV
END YW
END YX
END YY
END YZ
END ZA
END ZB
END ZC
END ZD
END ZE
END ZF
END ZG
END ZH
END ZI
END ZJ
END ZK
END ZL
END ZM
END ZN
END ZO
END ZP
END ZQ
END ZR
END ZS
END ZT
END ZU
END ZV
END ZW
END ZX
END ZY
END ZZ

Compare the two cells

comm -3 lib_cells.txt lef_cells.txt

- cells only in .lib
- cells only in .lef
- helps find missing or extra cells

Key Takeaways

- 1 Use basic Linux commands (ls, grep, awk, sort, comm) to quickly analyze LEF files.
- 2 Check file size, version, cells, pins, OBS, and compare with .lib.
- 3 These commands help you catch issues early and save debug time in physical design.

**Small Commands
Big Debugging Power!**

**cadence
Innovus**

Common LEF Errors and Debugging

Understand the issues, their impact, and how to debug them in innovus

Find the Issue
Fix the LEF
Get Back on Track!

1 Tech LEF is Missing

- Innovus cannot read layer, via, and technology information.
- You will see errors while reading cell LEF.
- PnR cannot start.

Typical Error Message

```
ERROR (ENCLE-001):  
Tech LEF must be read before  
cell LEF.
```

innovus> read_lef stdcell.lef

```
ERROR (ENCLE-001): No technology  
LEF is loaded.
```

Load Tech LEF first.

2 Cell LEF is Missing

- Innovus does not know the physical details of standard cells or macros.
- Cells are treated as black boxes with no size/pins.
- Placement and routing will fail.

Typical Error Message

```
WARNING (IMPSP-161):  
Cannot find LEF for cell 'NAND2_X1'.
```

innovus> read_db design.v

```
WARNING (IMPSP-161): Cannot find  
LEF for cell 'NAND2_X1'.
```

Provide correct cell LEF file.

3 Layer is Undefined

- Cell/tech LEF references a layer that is not defined in Tech LEF.
- Innovus cannot create shapes on that layer.
- PnR will stop due to missing layer info.

Typical Error Message

```
ERROR (LEF-003): Layer 'M7' is  
not defined.
```

innovus> read_lef stdcell.lef

```
ERROR (LEF-003): Layer 'M7' is  
not defined in technology LEF.
```

Check layer definitions in Tech LEF.

4 Cell Size is Incorrect

- If cell SIZE in LEF is wrong, placement grid and row alignment will be incorrect.
- Cells may overlap or have extra gaps.
- Leads to DRC violations and timing issues.

Correct Size

Incorrect Size (in LEF)

WARNING: cell size mismatch between LEF and reference.

Verify cell SIZE with foundry LEF.

5 Pin Location is Wrong

- Incorrect pin location causes routing failure or long detours.
- May lead to poor timing, DRC violations, or unconnected nets.

WARNING: (IMPSP-196): Pin location for 'Y' is outside the cell boundary.

Check pin coordinates in LEF.

6 OBS is Missing or Excessive

- If OBS is missing, router may place wires over blockages → DRC errors.
- If OBS is excessive, available routing area reduces → congestion and routing failure.

Missing OBS (Leads to DRC)

Excessive OBS (Leads to congestion)

Verify OBS in LEF.

7 SITE Mismatches

- If SITE in LEF does not match the one used in the flow, standard cells cannot be placed legally.
- Leads to placement errors and row creation issues.

Correct SITE

```
SITE CoreSite  
SIZE 0.240 BY 0.480 ;  
SYMMETRY X Y ;  
END CoreSite
```

Mismatched SITE

```
SITE WrongSite  
SIZE 0.300 BY 0.500 ;  
SYMMETRY X ;  
END WrongSite
```

Use the correct SITE from foundry LEF.

8 .lib, .v and .lef Cell Names Do Not Match

- If cell names are different across .lib, .v, and .lef, the tool cannot map logical cells to physical cells.
- Leads to missing cells, black boxes, or wrong cell usage.

File	Cell Name (Example)	Status
.v (Netlist)	NAND2_X1	✓
.lib (Timing)	NAND2_X1	✓
.lef (Physical)	NAND2_X1	✓
.lef (Physical)	NAND2X1	✗

WARNING (IMPSP-161): Cannot find LEF for cell 'NAND2_X1'.
WARNING (TIM-230): No timing for cell 'NAND2X1'.

Ensure consistent cell names in .lib, .v and .lef.

9 Impact on PnR

- These LEF errors can cause:
 - Failure in import
 - Placement failure or sub-optimal placement
 - Routing failures, DRC violations
 - Poor timing and increased congestion
 - Overall PnR flow may stop.

Read LEF/Design ✗ LEF errors stop the flow at early stage

Placement ✗ Wrong size/SITE/pins cause placement issues

Routing ✗ OBS/layer/pin issues cause routing problems

Signoff (DRC/TIMING) ✗ Leads to DRC and timing failures

Key Takeaways

- ✓ Always load Tech LEF before Cell LEF.
- ✓ Verify layers, SITE, cell sizes, pins, and OBS.
- ✓ Ensure consistent cell names in .lib, .v, and .lef.
- ✓ Check LEF using Innovus and Linux commands.
- ✓ Fix LEF issues early to avoid PnR failures.

Useful Innovus Commands for Debugging

```
head -20 file.lef # View file header (version, units)  
grep "LAYER" file.lef # Check layer definitions  
grep "SITE" file.lef # Check SITE information  
grep "MACRO" file.lef # List all cells/macros  
grep -A 5 "NAND2_X1" file.lef # View specific cell
```

```
reportSite # Show loaded sites  
reportLayers # Show layers from Tech LEF  
reportMacros # List all macros  
reportCell NAND2_X1 # Check cell details (size, pins, obs)  
checkDesign # Check for LEF/Lib/Netlist issues
```

**Clean LEF
→ Smooth PnR
→ Better Results!**

.lef

LEF File Structure – Part 1: Header & Technology Layers

Actual LEF Syntax | Example | Explanation

Understand LEF
Build a Better Chip!

Example: tech.lef (Technology LEF) – Part 1

```
1 VERSION 5.8 ;  
2 BUSBITCHARS "[ ]" ;  
3 DIVIDERCHAR "/" ;  
4 MANUFACTURINGGRID 0.001 ;  
5  
6 UNITS  
7 DATABASE MICRONS 1000 ;  
8 END UNITS  
9  
10 LAYER M1  
11 TYPE ROUTING ;  
12 DIRECTION HORIZONTAL ;  
13 PITCH 0.040 ;  
14 WIDTH 0.040 ;  
15 SPACING 0.040 ;  
16 AREA 0.040 ;  
17 MINWIDTH 0.040 ;  
18 MAXWIDTH 0.200 ;  
19 END M1  
20  
21 LAYER M2  
22 TYPE ROUTING ;  
23 DIRECTION VERTICAL ;  
24 PITCH 0.040 ;  
25 WIDTH 0.040 ;  
26 SPACING 0.040 ;  
27 AREA 0.040 ;  
28 MINWIDTH 0.040 ;  
29 MAXWIDTH 0.200 ;  
30 END M2  
31  
32 LAYER M3  
33 TYPE ROUTING ;  
34 DIRECTION HORIZONTAL ;  
35 PITCH 0.080 ;  
36 WIDTH 0.080 ;  
37 SPACING 0.080 ;  
38 END M3  
39  
40 ... (more layers like M4, M5, ..., VIA, etc.)
```

1 Header Information

- **VERSION**: Specifies the LEF version (e.g., 5.8).
- **BUSBITCHARS**: Defines the characters used for bus notation (e.g., []).
- **DIVIDERCHAR**: Defines the hierarchy divider (e.g., /).
- **MANUFACTURINGGRID**: Defines the manufacturing grid (minimum resolution) for all coordinates (in microns).

Example: MANUFACTURINGGRID 0.001 ; means 1 unit = 0.001 µm (1 nm).

2 Units Definition

- Defines the units used for all physical measurements in the LEF file.

UNITS

```
DATABASE MICRONS 1000 ;  
END UNITS
```

- Specifies that 1 LEF unit = 1/1000 micron (0.001 µm).
- All dimensions (width, pitch, spacing, etc.) are in these units.

3 Layer Definition

- Defines the physical and electrical properties of each metal layer used in the design.

LAYER M1

```
TYPE ROUTING ;  
DIRECTION HORIZONTAL ;  
PITCH 0.040 ;  
WIDTH 0.040 ;  
SPACING 0.040 ;  
AREA 0.040 ;  
MINWIDTH 0.040 ;  
MAXWIDTH 0.200 ;  
END M1
```

- **TYPE**: Type of layer (ROUTING, CUT, MASTERSLICE, etc.).
- **DIRECTION**: Preferred routing direction (HORIZONTAL/VERTICAL).
- **PITCH**: Recommended routing pitch.
- **WIDTH**: Default/nominal routing width.
- **SPACING**: Minimum spacing between wires.
- **AREA**: Minimum area for shapes.
- **MINWIDTH**: Minimum allowed width.
- **MAXWIDTH**: Maximum allowed width.

3D View: Metal Layers in a Chip (Simplified)

→ M3 (Horizontal)
→ M2 (Vertical)
→ M1 (Horizontal)
→ Substrate

Key Takeaways (Part 1)

- ✓ Header defines version and grid information.
- ✓ Units set the measurement scale.
- ✓ Each metal layer has defined routing rules.
- ✓ Tech LEF provides this technology information to the PnR tool (e.g., Innovus).
- ✓ This is only Part 1. Next, we will see SITE, VIAS, and other technology definitions in Part 2.

.lef

LEF File Structure – Part 2: SITE, VIAS and MACRO Definition

Actual LEF Syntax | Example | Explanation

Libraries Define the Chip
Build the Future!

Example: stdcell.lef (Cell LEF) – Part 2

```

41 SITE CoreSite
42 SIZE 0.240 BY 0.480 ;
43 SYMMETRY X Y ;
44 CLASS CORE ;
45 END CoreSite

47 VIA VIA12
48 LAYER M1 ;
49 RECT 0.000 0.000 0.040 0.040 ;
50 END
51 LAYER M2 ;
52 RECT 0.000 0.000 0.040 0.040 ;
53 END
54 RESISTANCE 0.02 ;
55 CAPACITANCE 0.001 ;
56 END VIA12

58 MACRO NAND2_X1
59 CLASS CORE ;
60 FOREIGN NAND2_X1 0 0 ;
61 ORIGIN 0 0 ;
62 SIZE 1.20 BY 2.00 ;
63 SITE CoreSite ;
64 SYMMETRY X Y ;
65 PIN A
66 DIRECTION INPUT ;
67 USE SIGNAL ;
68 PORT
69 LAYER M1 ;
70 RECT 0.10 1.80 0.30 2.00 ;
71 END
72 END A
73 ... (other pins) ...
74 END NAND2_X1

```

1 SITE Definition

- Defines the standard cell placement site (row height, width, symmetry).
- Used by PnR tool to create rows and place standard cells legally.
- All standard cells in a library use the same site (e.g., CoreSite).

SITE CoreSite
 SIZE 0.240 BY 0.480 ;
 SYMMETRY X Y ;
 CLASS CORE ;
 END CoreSite

2 VIA Definition

- Defines the via used to connect two metal layers.
- Specifies the geometry (RECT), resistance and capacitance.
- Used by the PnR tool for routing and RC extraction.

VIA VIA12
 LAYER M1 ;
 RECT 0.000 0.000 0.040 0.040 ;
 END
 LAYER M2 ;
 RECT 0.000 0.000 0.040 0.040 ;
 END
 RESISTANCE 0.02 ;
 CAPACITANCE 0.001 ;
 END VIA12

3 Macro Definition (Standard Cell)

- Defines each standard cell in the library.
- Includes size, site, pins, symmetry, and other physical information.
- Used by tool for placement, legalization, routing and DRC.

MACRO NAND2_X1
 CLASS CORE ;
 SIZE 1.20 BY 2.00 ;
 SITE CoreSite ;
 SYMMETRY X Y ;
 ... (pins, obs, etc.) ...
 END NAND2_X1

Key Takeaways (Part 2)

- ✓ SITE defines the placement grid for standard cells.
- ✓ VIA defines inter-layer connections with geometry and RC.
- ✓ MACRO defines each standard cell with size, site, pins, etc.
- ✓ This information is essential for legal placement, routing and accurate parasitic extraction.
- ✓ Next (Part 3) we will see PIN details, OBS (obstructions), and end of LEF.

Correct LEF = Smooth PnR = Better Chips!

LEF File Structure – Part 3 : PINS, OBS and End of Cell

Actual LEF Syntax | Example | Explanation

Example: stdcell.lef (Cell LEF) – Part 3

```

78 MACRO NAND2_X1
79 ... (previous content from Part 2)
80
81 PIN A
82 DIRECTION INPUT ;
83 USE SIGNAL ;
84 PORT
85 LAYER M1 ;
86 RECT 0.10 0.30 0.30 0.50 ;
87 END
88 END A
89
90 PIN Y
91 DIRECTION OUTPUT ;
92 USE SIGNAL ;
93 PORT
94 LAYER M1 ;
95 RECT 0.10 0.30 1.30 0.50 ;
96 END
97 END Y
98
99 PIN VDD
100 DIRECTION INOUT ;
101 USE POWER ;
102 PORT
103 LAYER M1 ;
104 RECT 0.10 0.70 1.30 2.00 ;
105 END
106 END VDD
107
108 PIN VSS
109 DIRECTION INOUT ;
110 USE GROUND ;
111 PORT
112 LAYER M1 ;
113 RECT 0.10 0.00 1.10 0.30 ;
114 END
115 END VSS
116
117 OBS
118 LAYER M1 ;
119 RECT 0.40 0.80 0.80 1.20 ;
120 LAYER M2 ;
121 RECT 0.50 0.90 0.70 1.10 ;
122 END
123
124 END NAND2_X1

```

1 PIN Definition (Input/Output)

- Defines the pin name, direction, use (signal/power/ground), and the physical location (PORT with layer and geometry).
- Pins are used by the PnR tool for placement, routing and timing analysis.

PIN A
 DIRECTION INPUT ;
 USE SIGNAL ;
 PORT
 LAYER M1 ;
 RECT 0.10 0.30 0.30 0.50 ;
 END
 END A

2 PIN Definition (Output)

- Output pins carry the signal out of the cell.
- The location helps the tool in timing-driven placement and routing.

PIN Y
 DIRECTION OUTPUT ;
 USE SIGNAL ;
 PORT
 LAYER M1 ;
 RECT 1.10 0.30 1.30 0.50 ;
 END
 END Y

3 Power and Ground Pins

- Power pins (VDD) use USE POWER and ground pins (VSS) use USE GROUND.
- Usually placed on top and bottom of the cell for easy power rail connection.

PIN VDD
 DIRECTION INOUT ;
 USE POWER ;
 PORT
 LAYER M1 ;
 RECT 0.10 0.70 1.10 2.00 ;
 END
 END VDD

PIN VSS
 DIRECTION INOUT ;
 USE GROUND ;
 PORT
 LAYER M1 ;
 RECT 0.10 0.00 1.10 0.30 ;
 END
 END VSS

4 OBS (Obstruction)

- Defines the blocked regions inside the cell where routing is not allowed (e.g., due to internal diffusion, implants or blockage).
- Helps the PnR tool avoid illegal routing and DRC errors.

OBS
 LAYER M1 ;
 RECT 0.40 0.80 0.80 1.20 ;
 LAYER M2 ;
 RECT 0.50 0.90 0.70 1.10 ;
 END

5 End of Cell

- END <cell_name> marks the end of the macro definition.

END NAND2_X1 Completes the definition of this cell in the LEF file.

Key Takeaways (Part 3)

- ✓ Pins define connectivity and physical access.
- ✓ Power/Ground pins enable power planning.
- ✓ OBS prevents illegal routing.
- ✓ Correct pin and OBS information is essential for clean PnR.
- ✓ END statement completes the cell definition.

Good LEF = Clean Design = Smooth PnR!

